

ECE445 Final Presentation

Team #29 - Mimicry Stage Lighting Control System

Team #29

RUIQI LI - ECE

HAOZHE CHEN - ECE

ZHAOHUA YANG - EE

ANQI TAN - ME

Sponsor: Timothy Lee

May 20, 2022



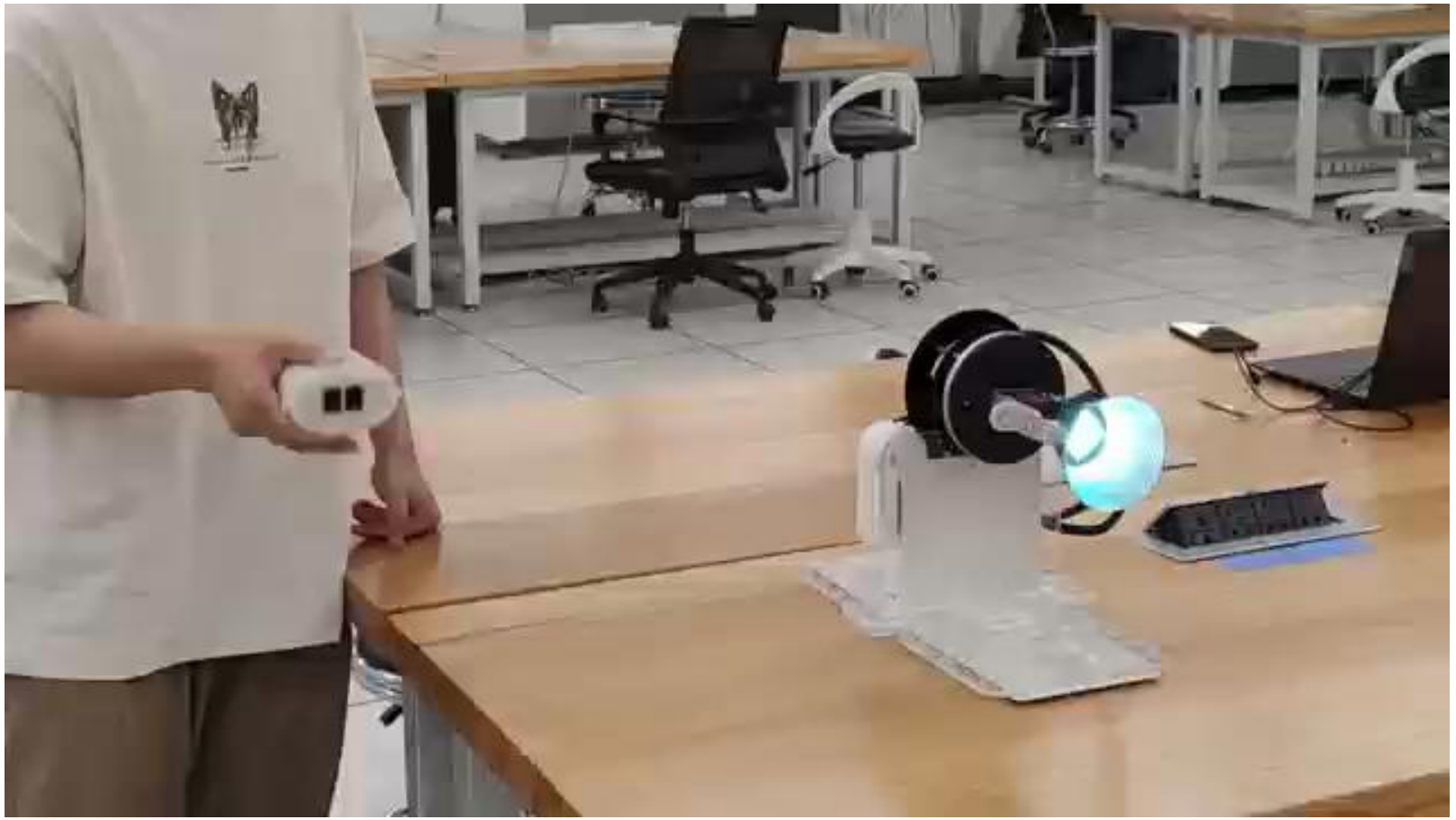
1 Introduction

Introduction – Background & Problem

- Background
  - Stage Lighting Control System

- Problems for smaller and amateur circumstances
  - It's too Unwieldy
  - It's difficult to operate
  - It lacks liveness

Our Solution: Mimicry Stage Lighting Control System



Highlevel Requirement

- Precision
- Portability
- User-friendliness



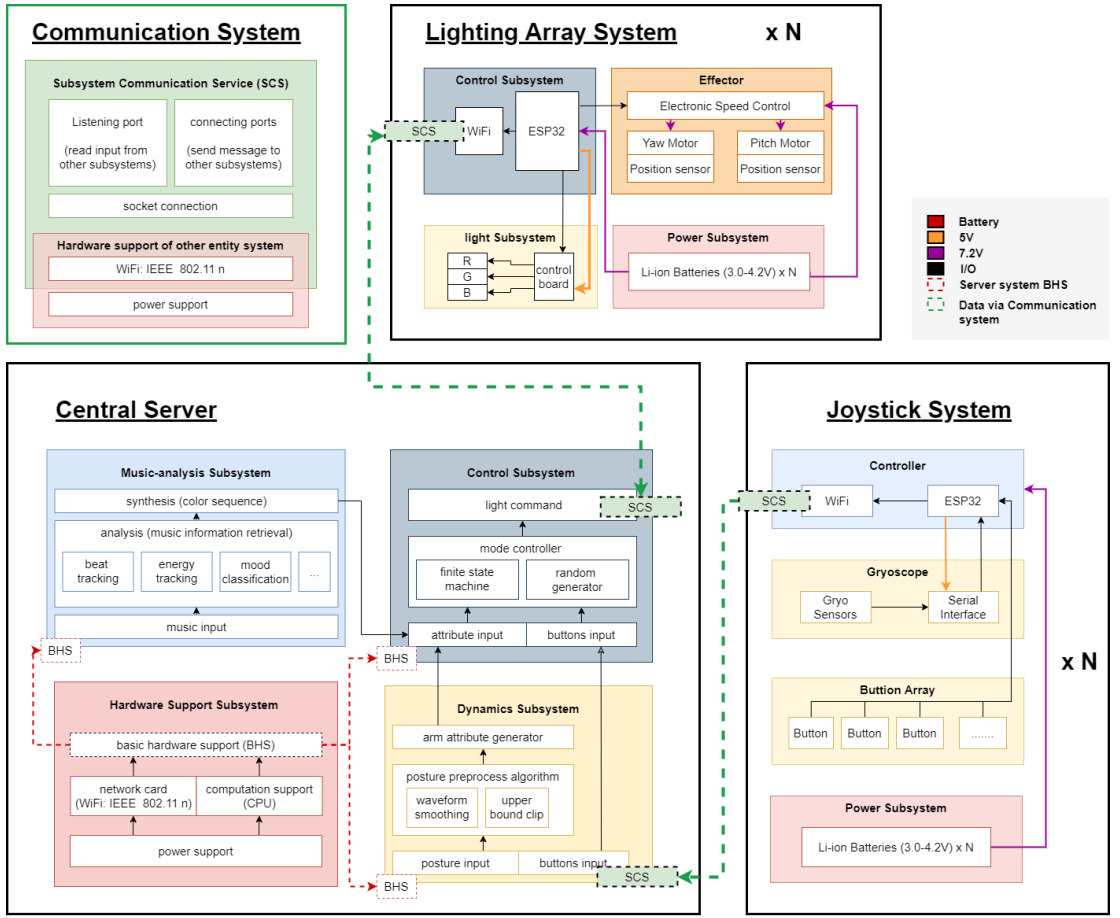
2 Design

Block Diagram

Four systems,

in order of dataflow:

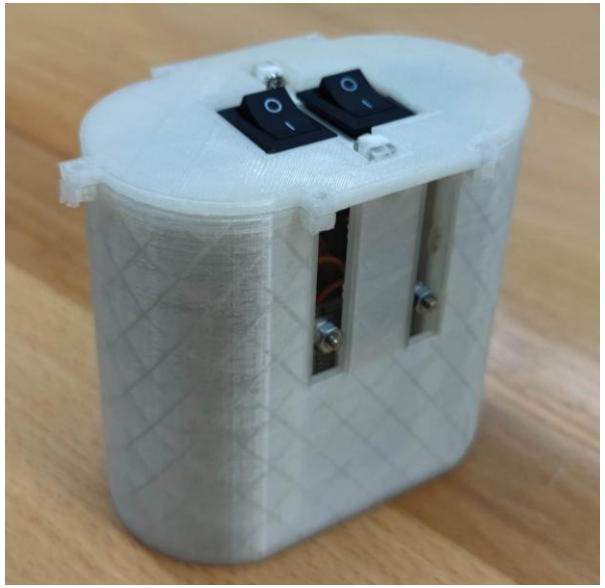
- Joystick system
- Communication system
- Central server
  - Music Analysis Subsystem
  - Dynamic Subsystem
  - Control Subsystem
- Lighting array system

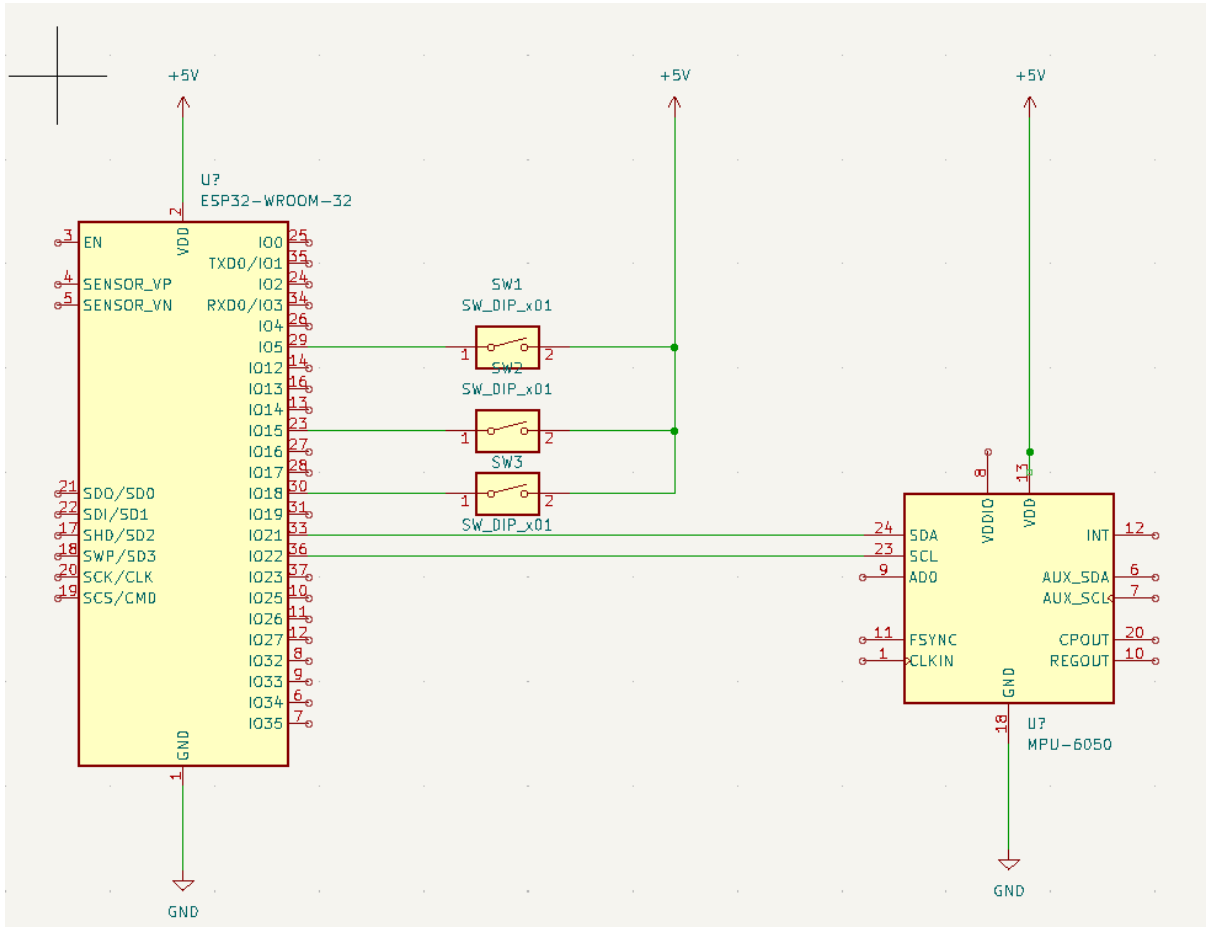


Joystick System - Design

A controller that DJ hold to control the direction of light

- ESP32
- MPU9250
- Buttons





Joystick System - Design



ESP32:

- Dual-core MCU
- Wireless communication
- GPIO Matrix

MPU9250:

- Posture sensor

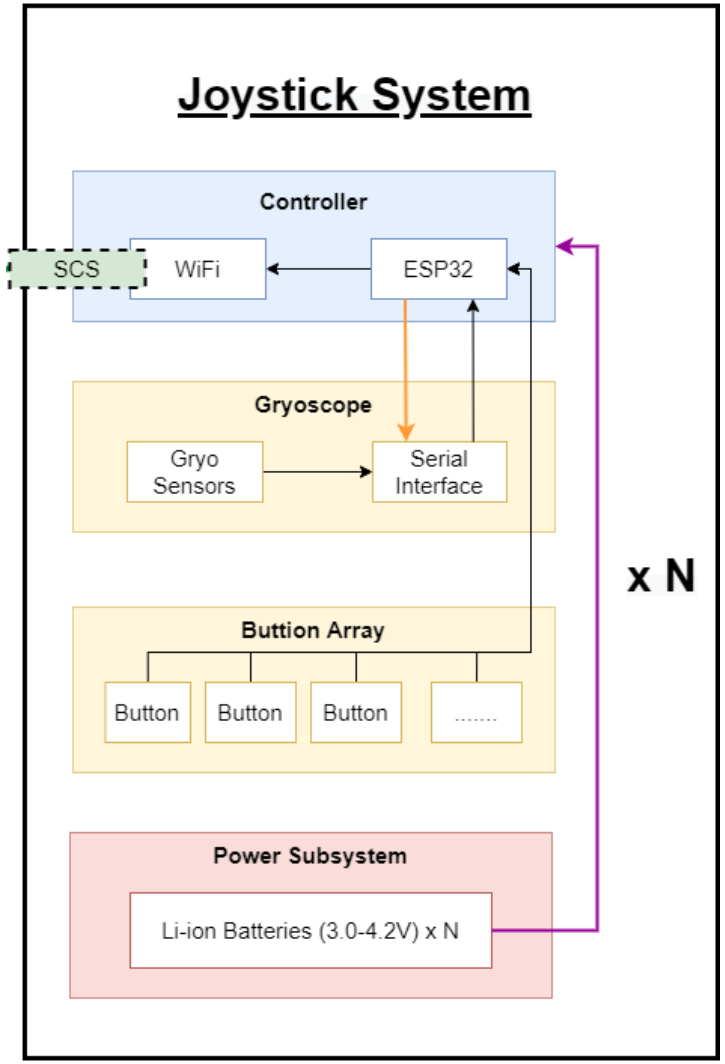
Button Array:

- Need to debounce

Battery:

Type: 18650

- Voltage: 3.0-4.2V.



Joystick System - R&V & Results

Requirements

- Preprocess the original signals to avoid inherent error of sensor.
- Communicate wirelessly with other devices.
- Detect the direction in real time.
- Continuously work for a long time

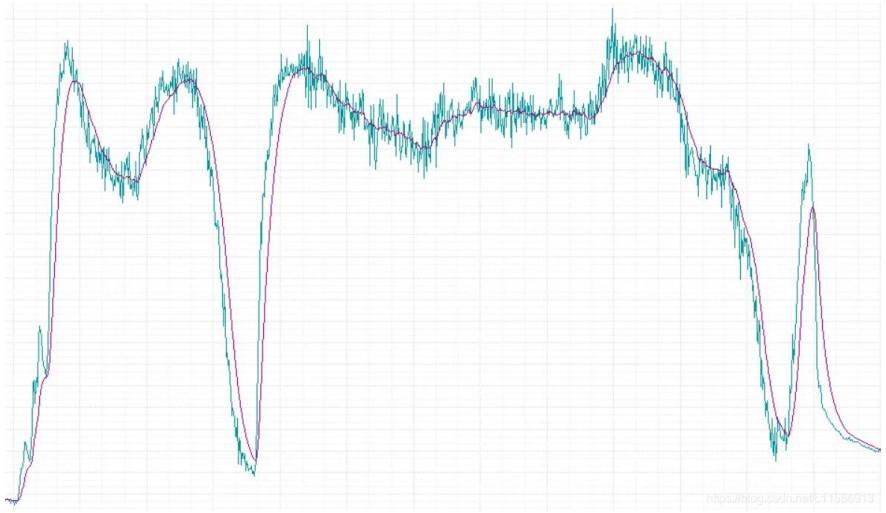
Verification & Results

- **ESP32** is strong enough to receive and preprocess the data and supports wireless communication.
- **MPU9250** can detect acceleration, angular velocity and magnetic value
- The average working current is **100 mA**, which support continuously **working for 15h.**

Joystick System – Challenge: Posture Detection Error

The data from MPU9250 is risk of noise, missing and singularity.

- Circuit interrupt.
- Read/Write fast or slow
- Intrinsic flaw of I2C communication.

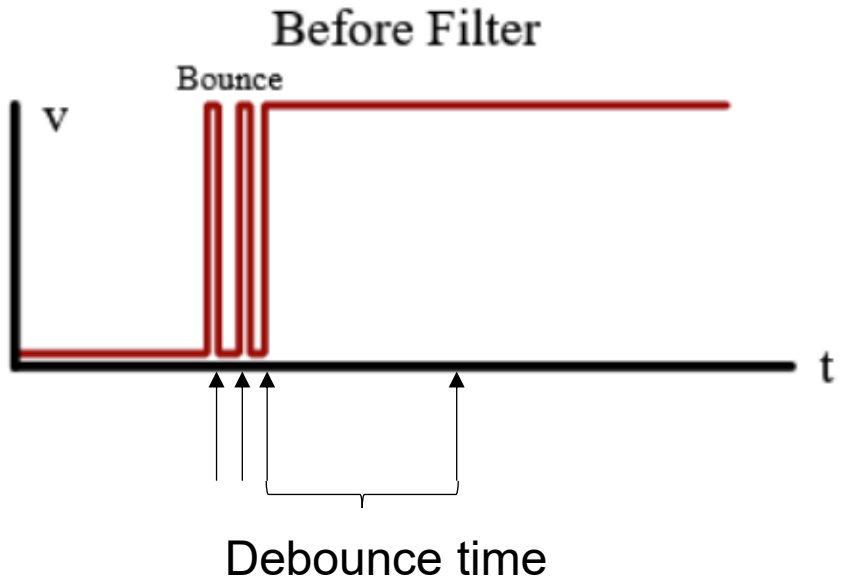


Kalman filter can give the best estimation at every time

Joystick System – Challenge: Button Debounce

Due to the pandemic, we do not have buttons that can debounce automatically. Therefore, we need to debounce by software

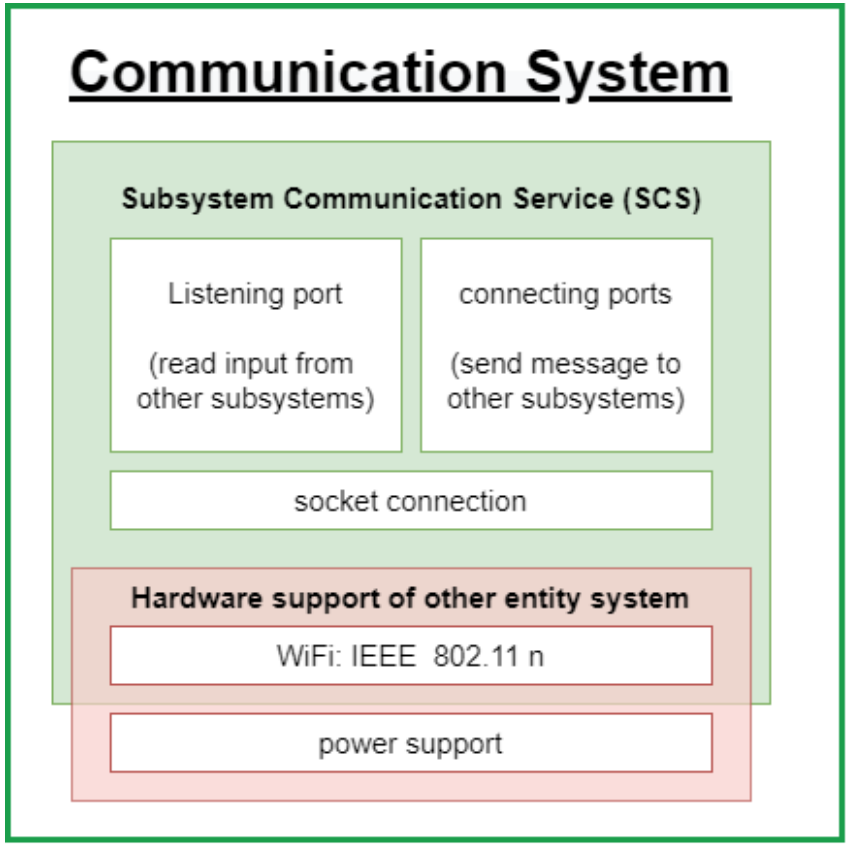
- Read after signal stays still for some time.



Communication System - Design

An abstract software system handles all the wireless connections

- WiFi environment and the devices
- Reliable connection
  - Disconnection detection
  - Auto re-connection
- Multi-device connection
- Low delay



Communication System – R&V & Results

Requirements

--- Environment

- The router should support **IEEE 802.11n** protocol. (**WiFi**)
- The wireless **signal should be strong** enough for maintaining stable communication.

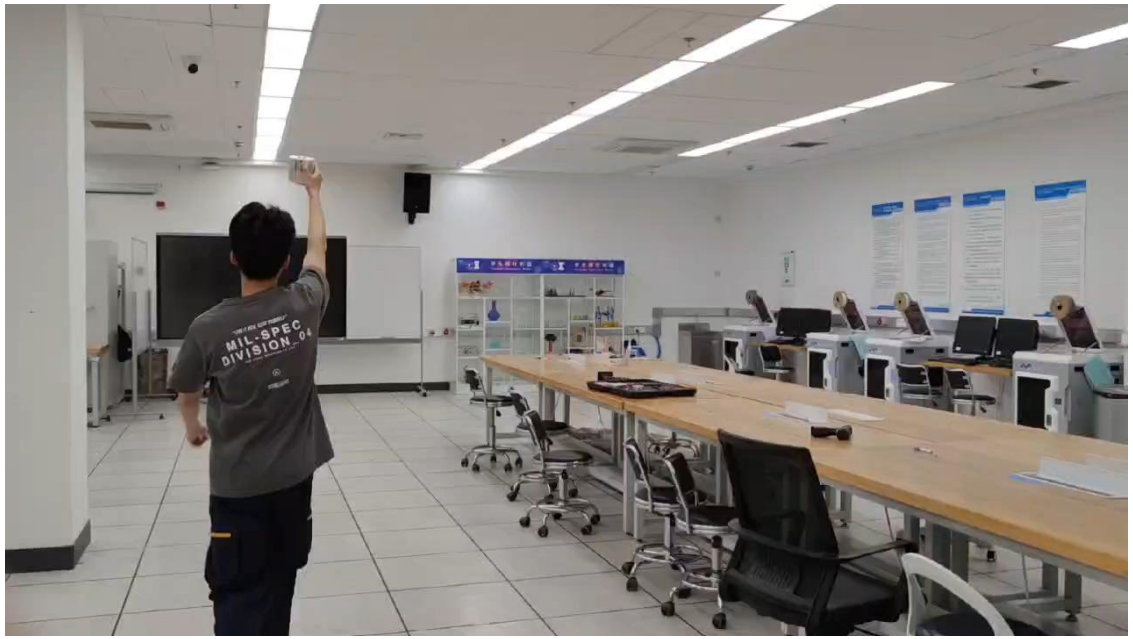
Verification & Results

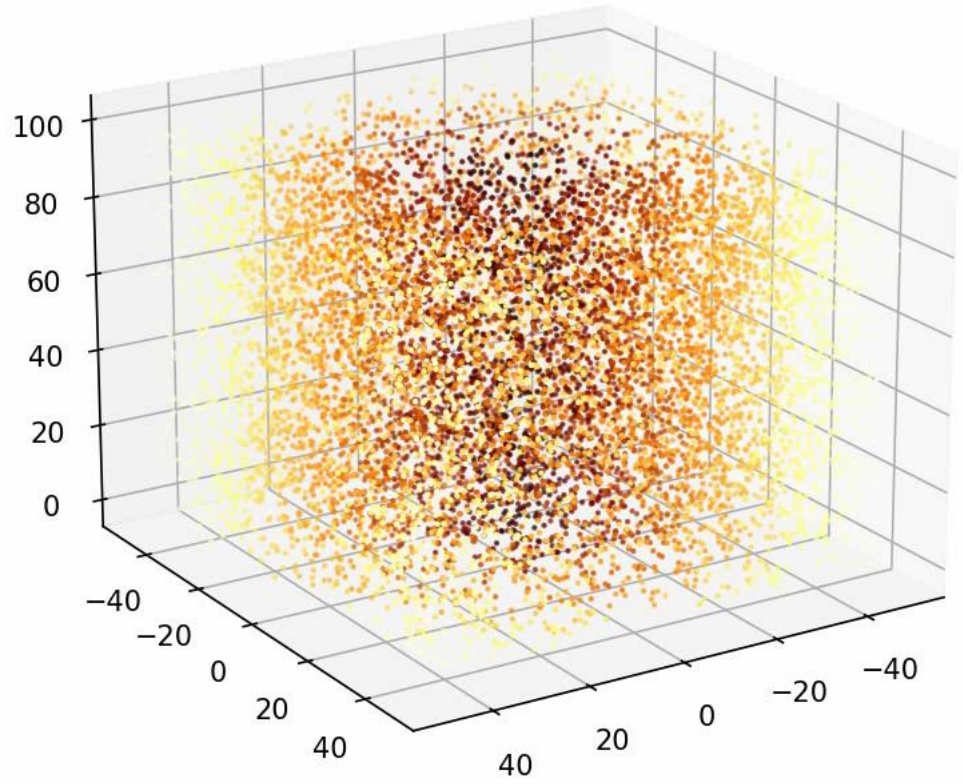
- The WiFi environment should be easily established by common devices.
- The signal strength is detected and verified with the ESP32 board.
- The result shows **the signal is strong enough to maintain reliable connections.**

Modification

- No modification

Communication System – R&V & Results





Communication System – R&V & Results

Requirements

--- Stability

- The system should support **at least 32 connections**.
- The connection delay should be **limited to 30ms**.
- The connection failure should be detected in 100ms and automatically repair it.

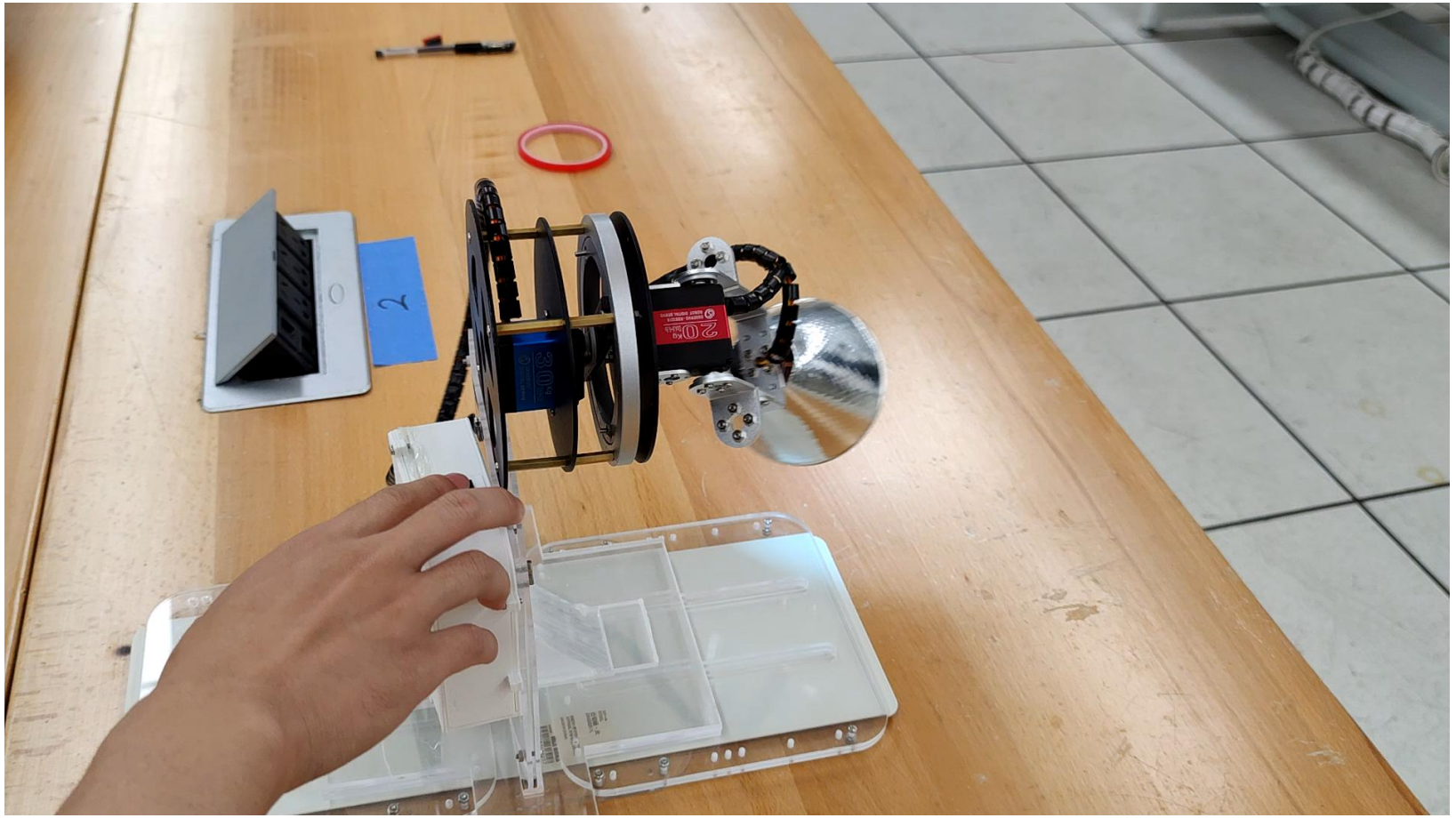
Verification & Results

- The **socket protocol** is used, which support **16M** connections.
- The delay is tested with timestamp label and delay is **limited to 20ms.**
- The failure connection timeout is programed to 100ms. **A hot plug reconnection is realized**.

Modification

- No modification.

Communication System – R&V & Results



Communication System – Challenge

The delay between devices has been reduced **from 30s to 20ms**.

Main factors of delay

- Remain data in the queue
- Default socket configuration:
  - Large buffer
  - Low sending frequency

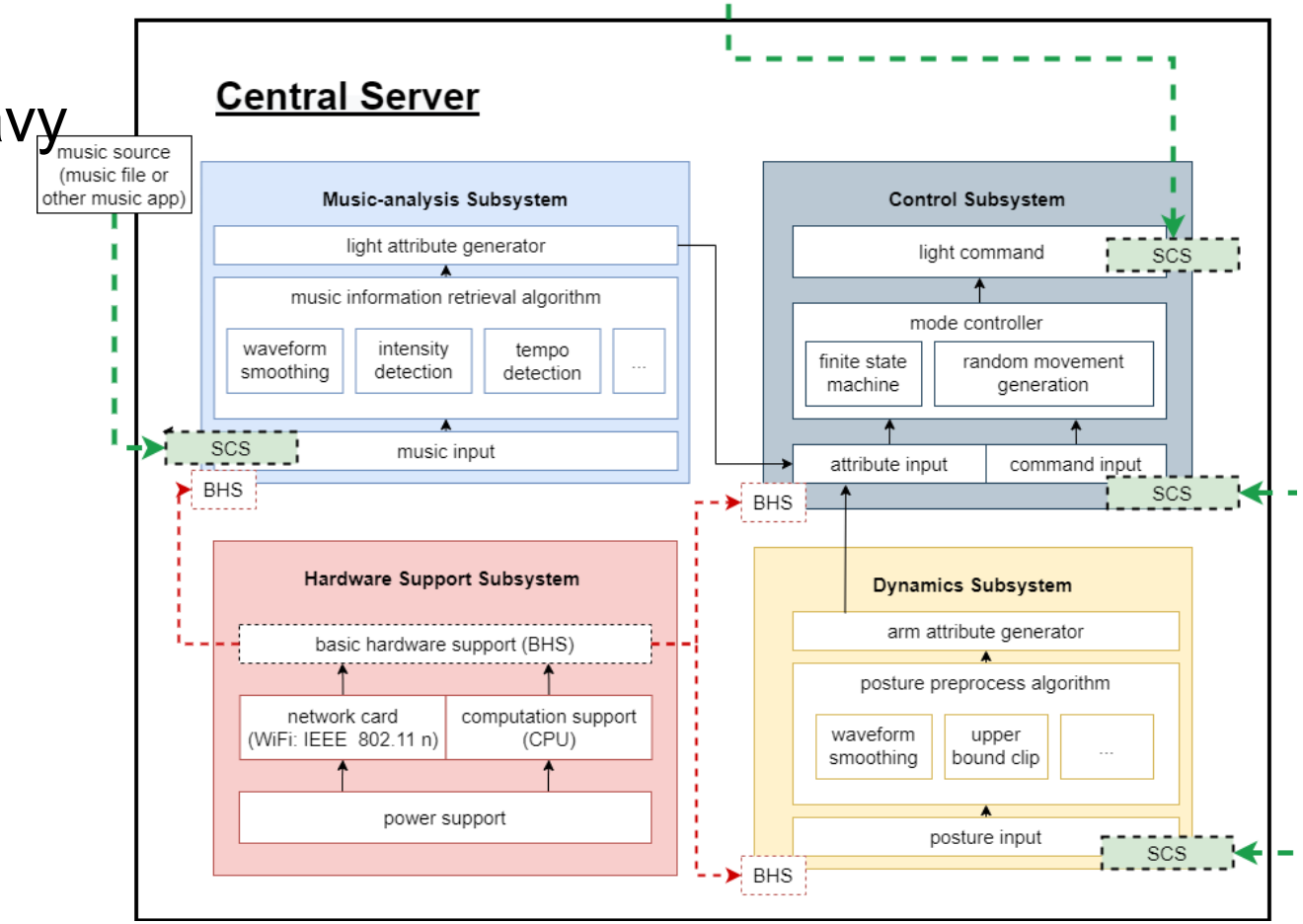
Solutions

- **Attaching timestamps** to message to guarantee only in-time message will be processed.
- **Using shared data with lock** instead of queue to avoid redundant data.
- Changing socket sending frequency to 10ms and setting buffer size to 3 times of message size.

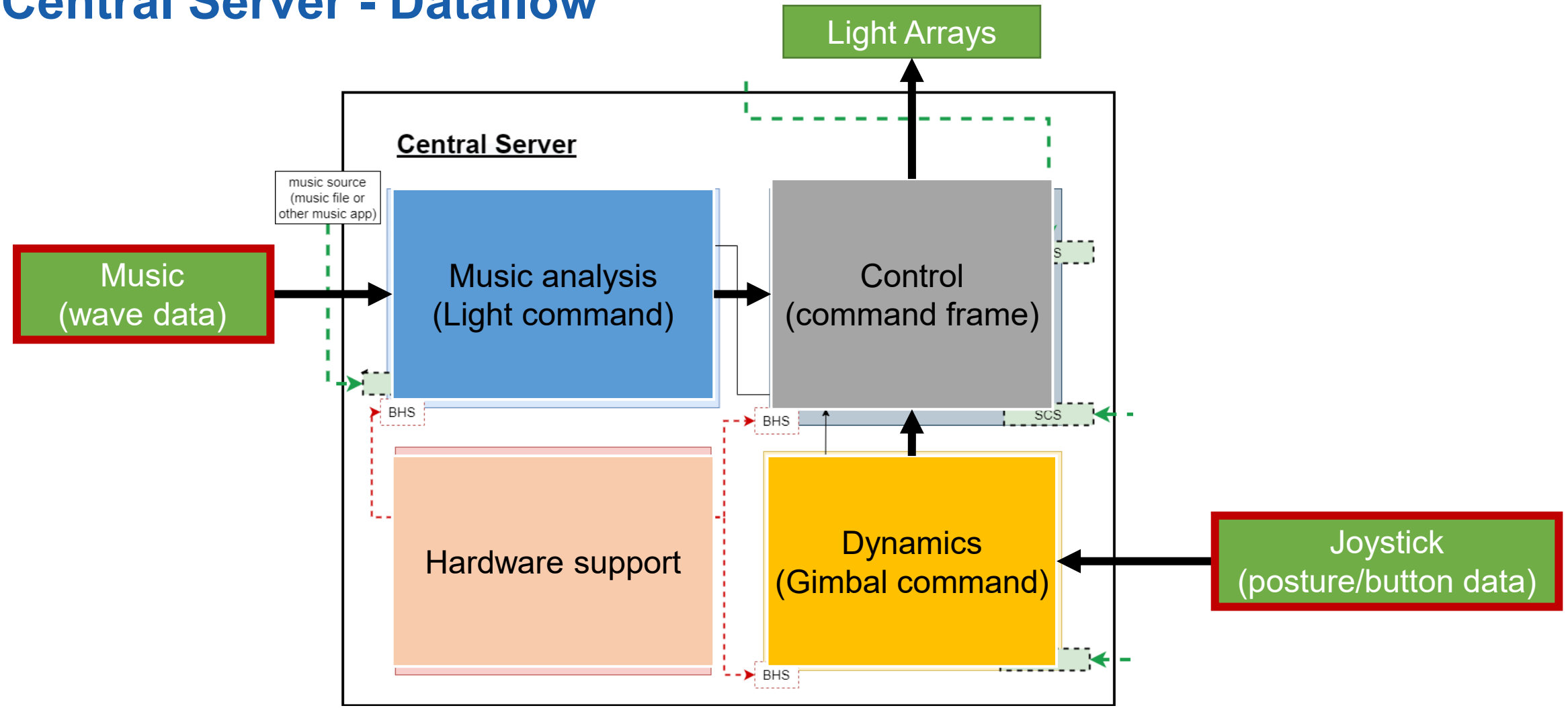
Central Server

A central computation platform handles all the computational heavy process

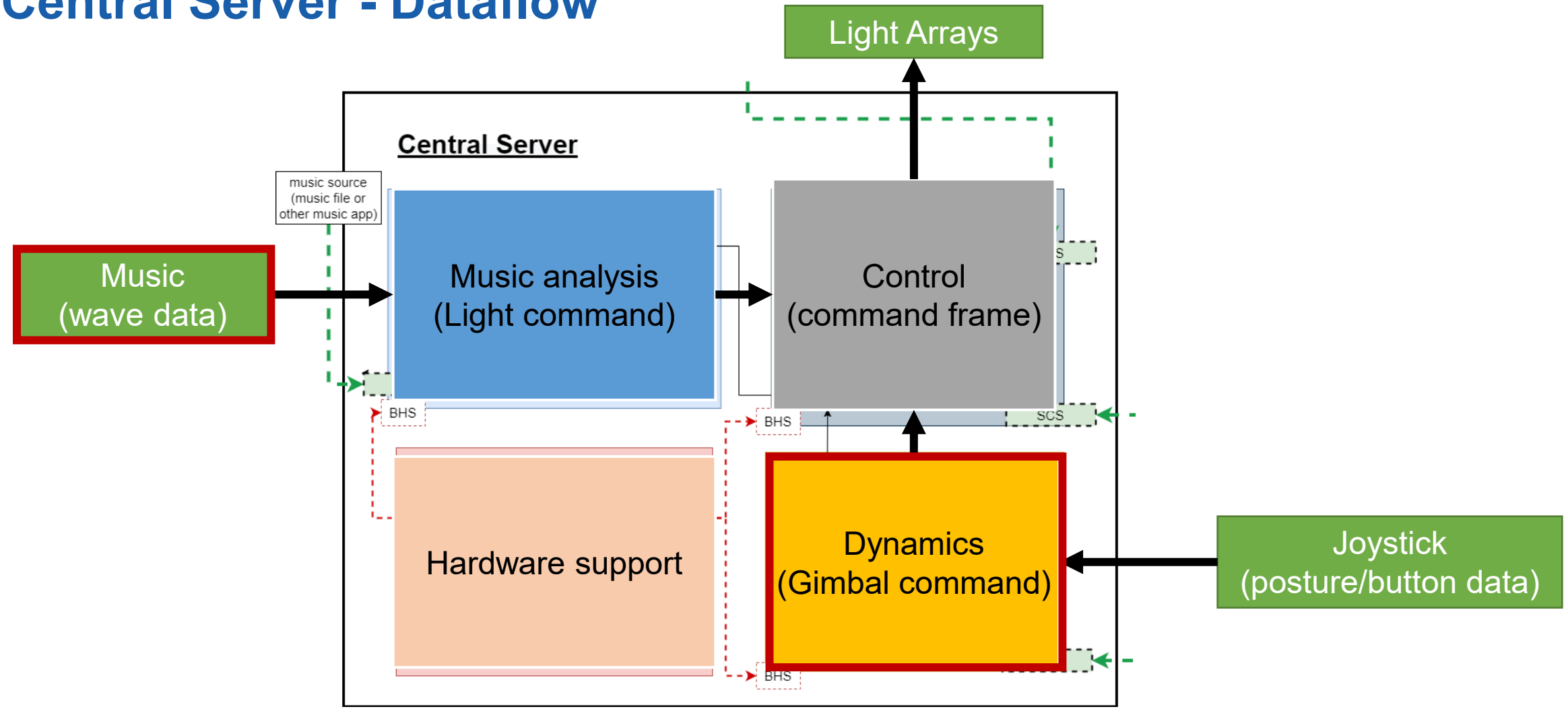
- Dynamic Subsystem
- Control Subsystem
- Music Analysis Subsystem
- Hardware Support Subsystem
  - Power support
  - Network card support WiFi



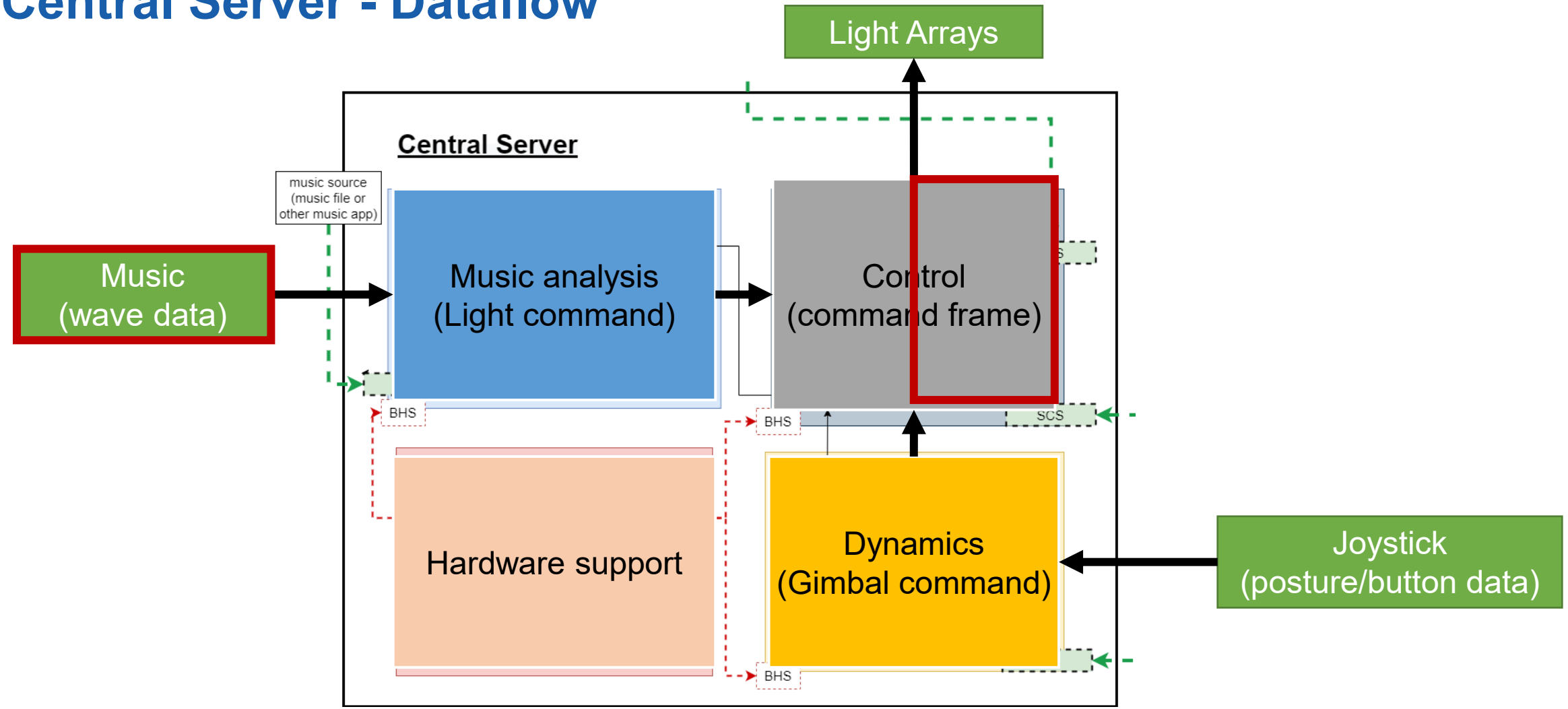
Central Server - Dataflow



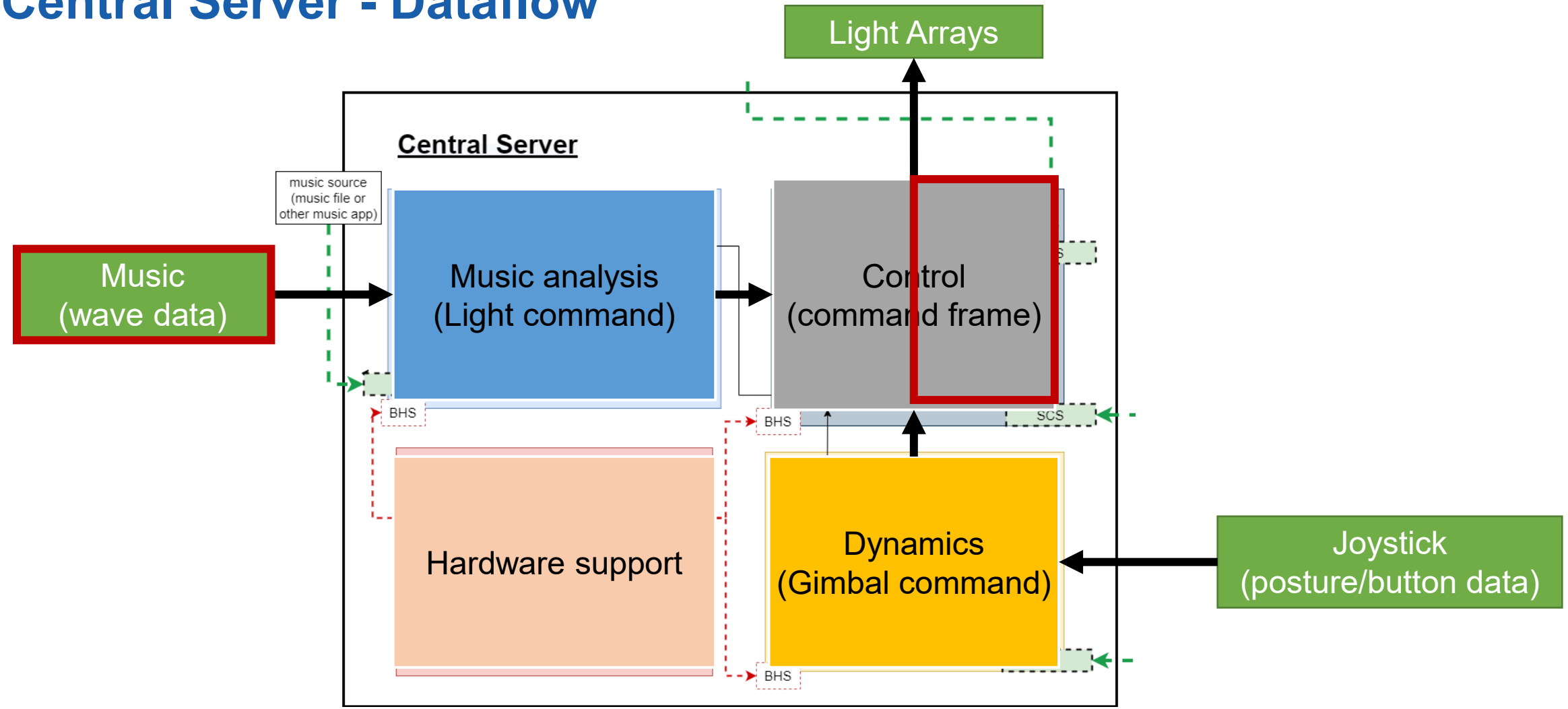
Central Server - Dataflow



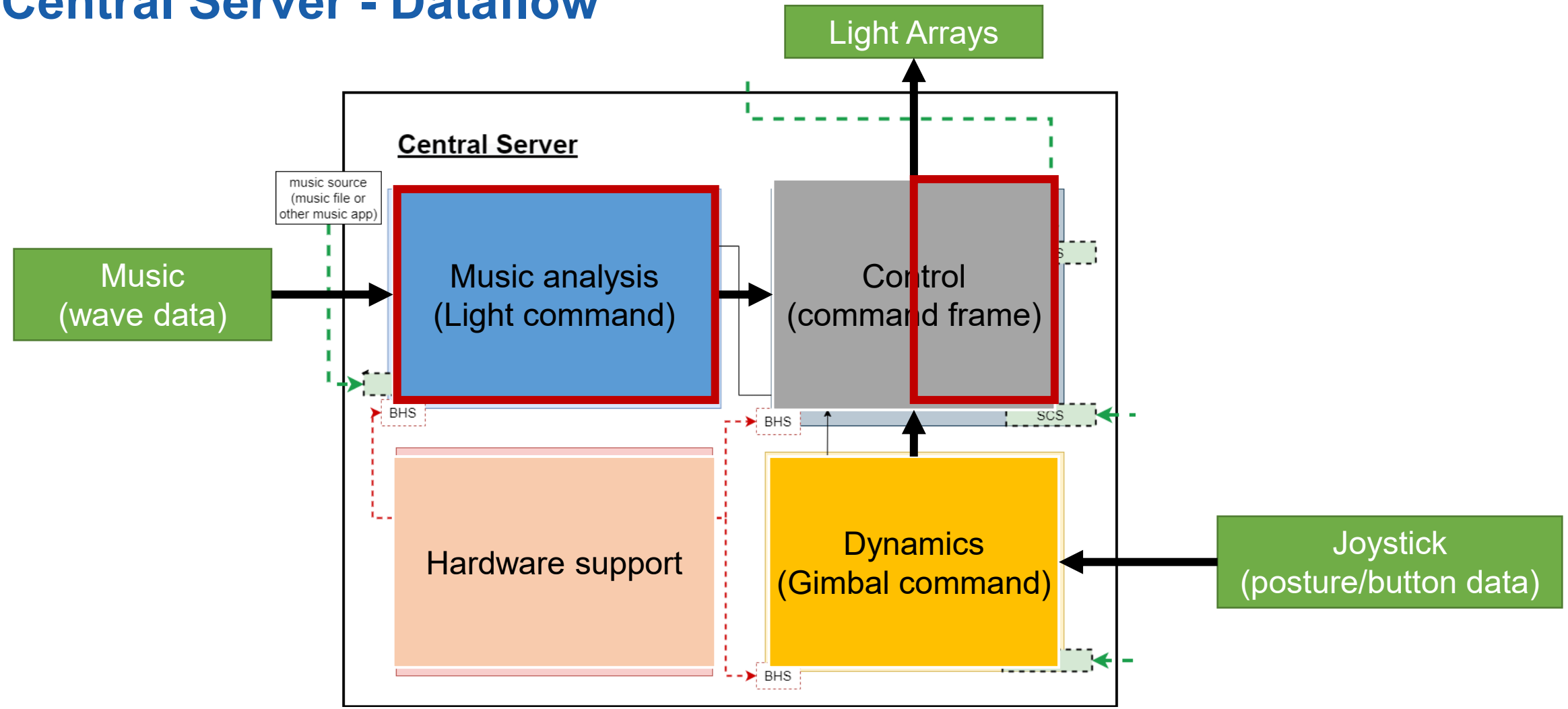
Central Server - Dataflow



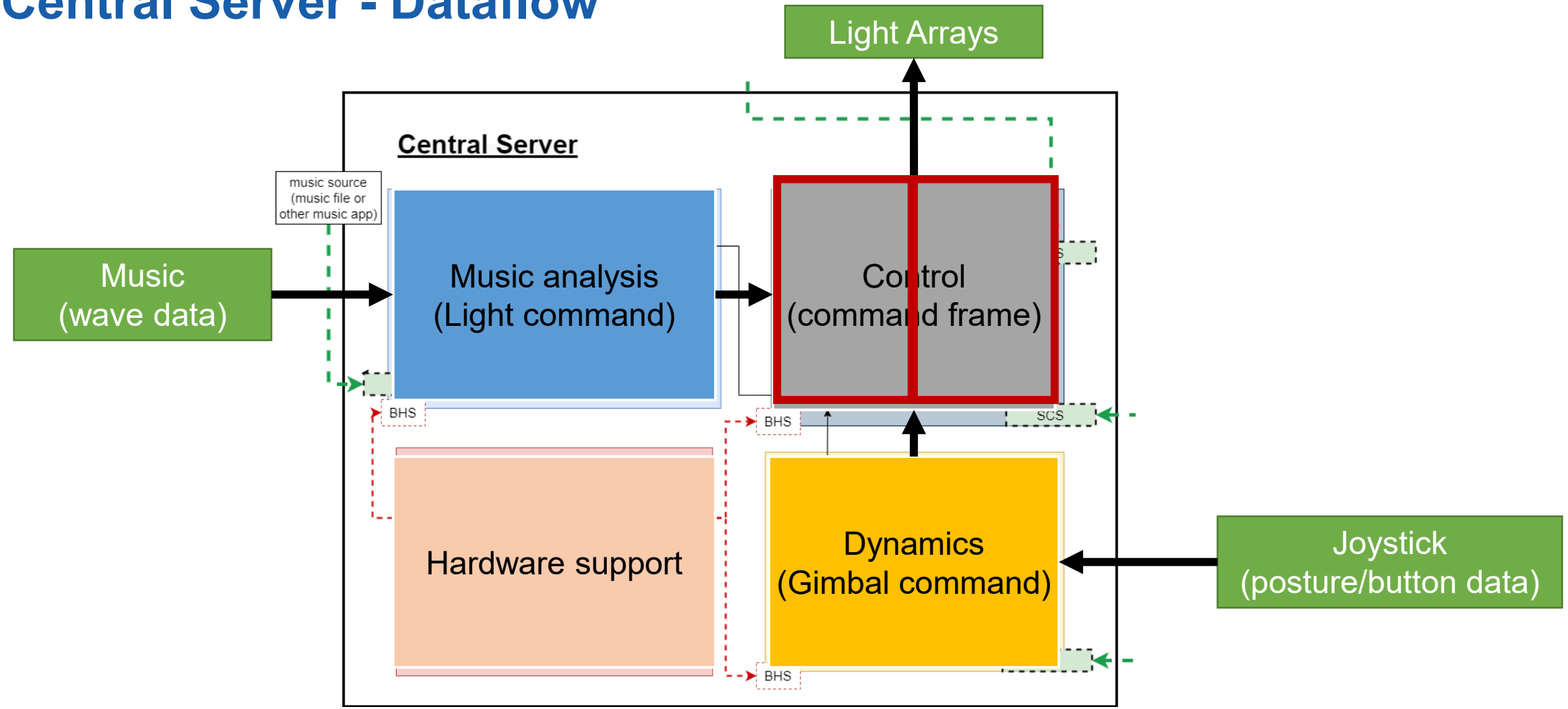
Central Server - Dataflow



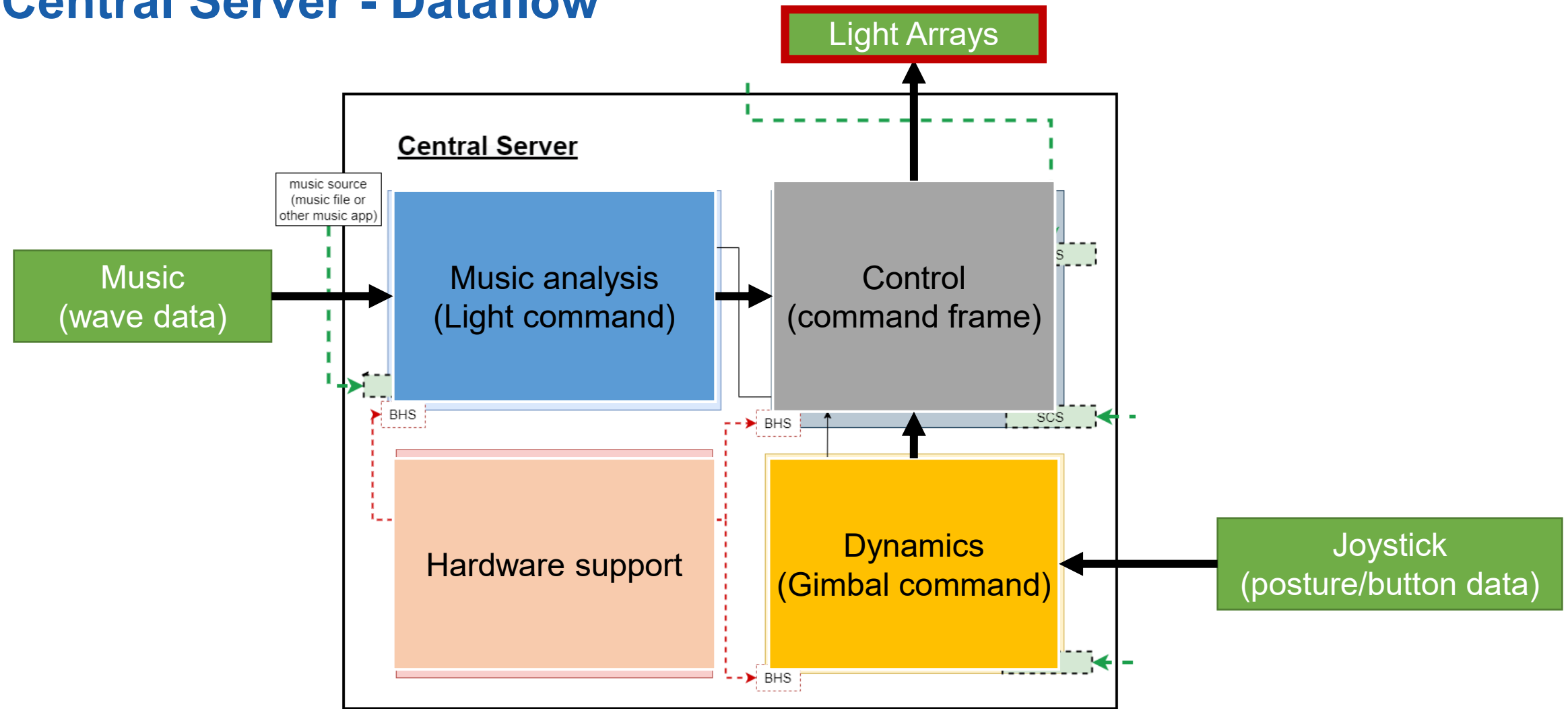
Central Server - Dataflow



Central Server - Dataflow



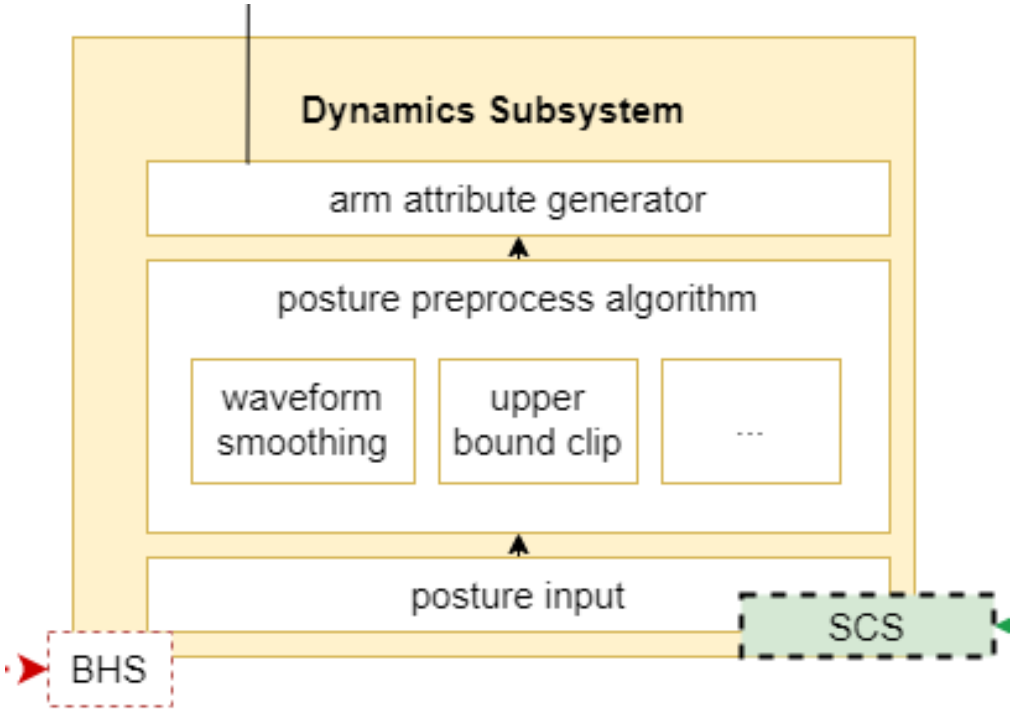
Central Server - Dataflow



Dynamics Subsystem - Design

Preprocess and transmit posture data from joystick

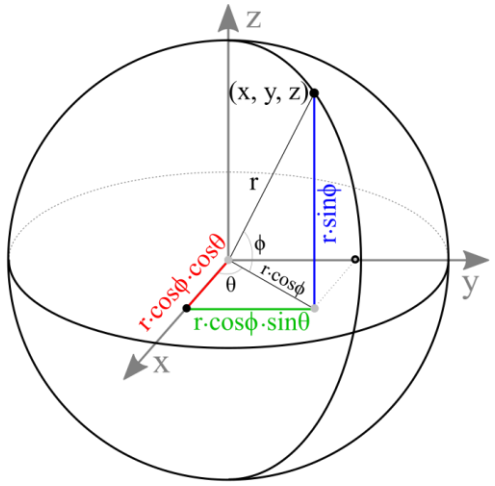
- Button data transmission
- Spatial Remap
- Temporal Remap



Dynamics Subsystem – Challenge: Spatial Remap

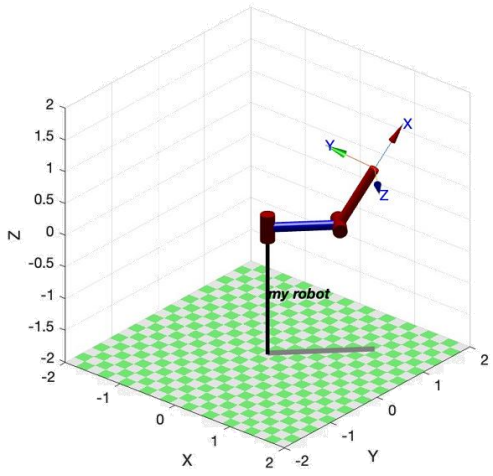
The coordinate system conversion

- From cartesian coordinate system to **Euler angles**
- (implemented in joystick)



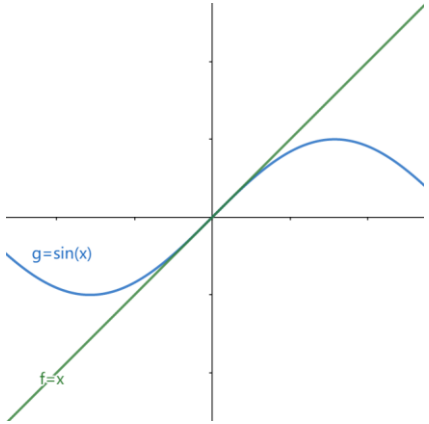
The inverse kinematics

- Turn the **3D** posture data to the **2D** servo command.



The non-linear upper bound clip

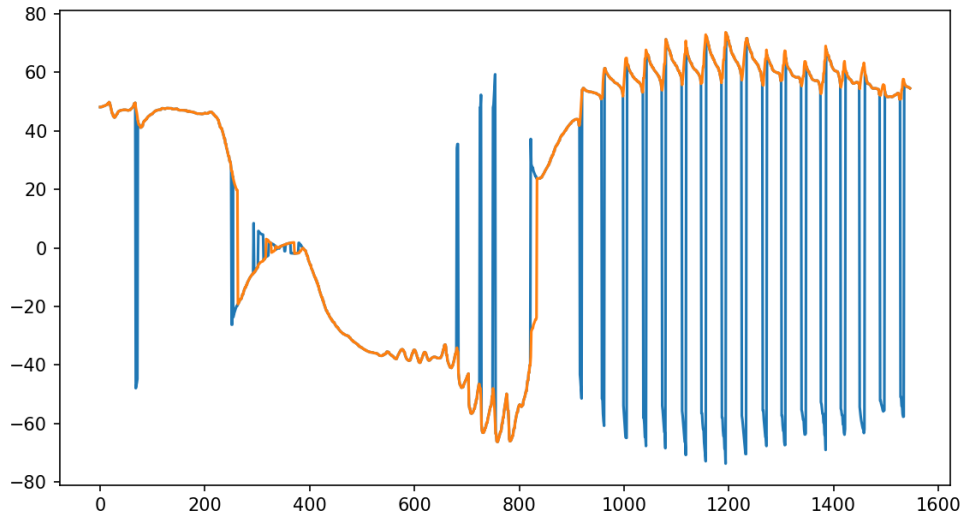
- Restrict the max angle to adapt the physical limitation of mechanical structure (**scaled *sin***)



Dynamics Subsystem – Challenge: Temporal Remap

- The pitch/roll angle was derived from acceleration, severe oscillation will cause sudden reversed sample data.
- A **sign decision rule based on causal convolution** is implemented.

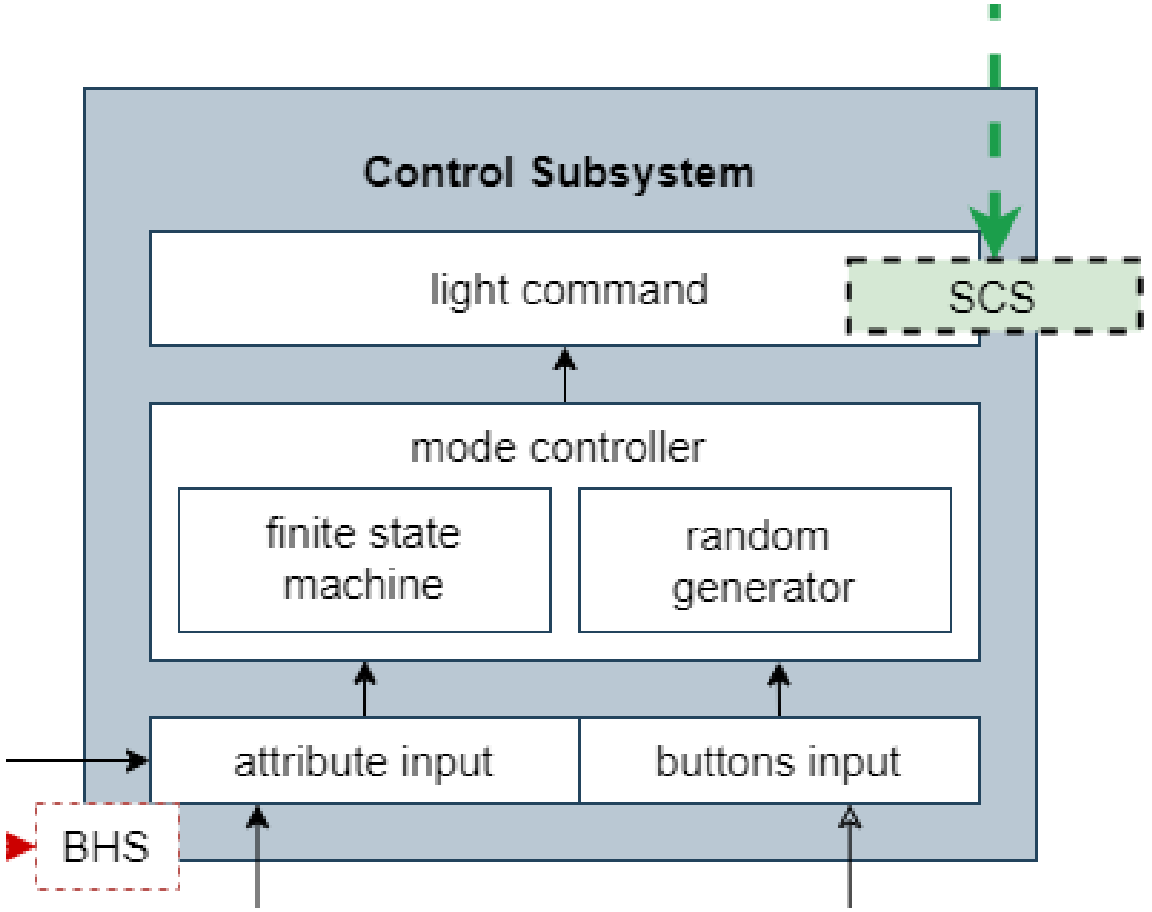
$$sign(X_T) = sign(\sum_{t=0}^{t=20} X_{T-t} * \delta(t))$$



Control Subsystem - Design

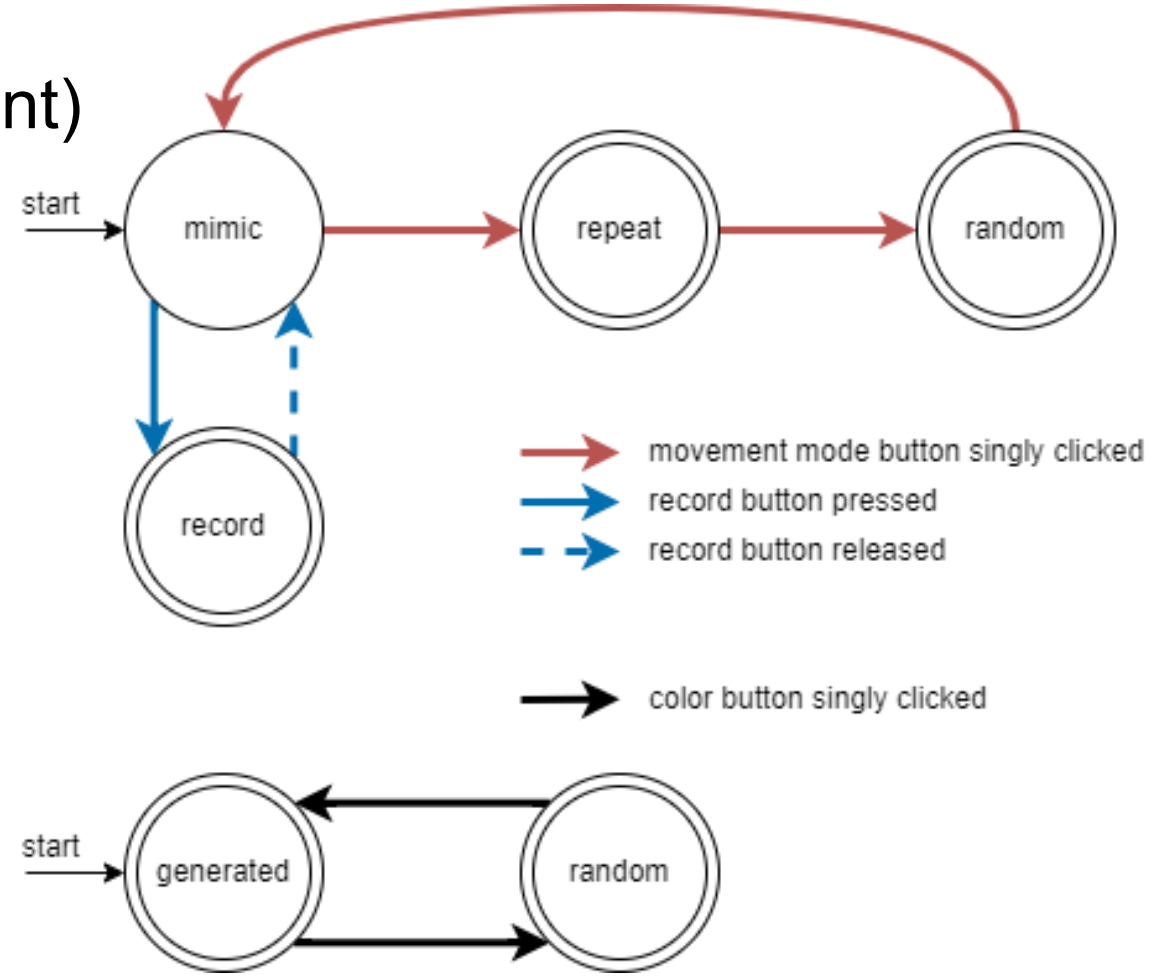
- Works as a mode controller and command combinator

- Random Generator
  - Generate random color and movement command according to the music feature
- Finite State Machines
  - Used the state to filter the light color and movement

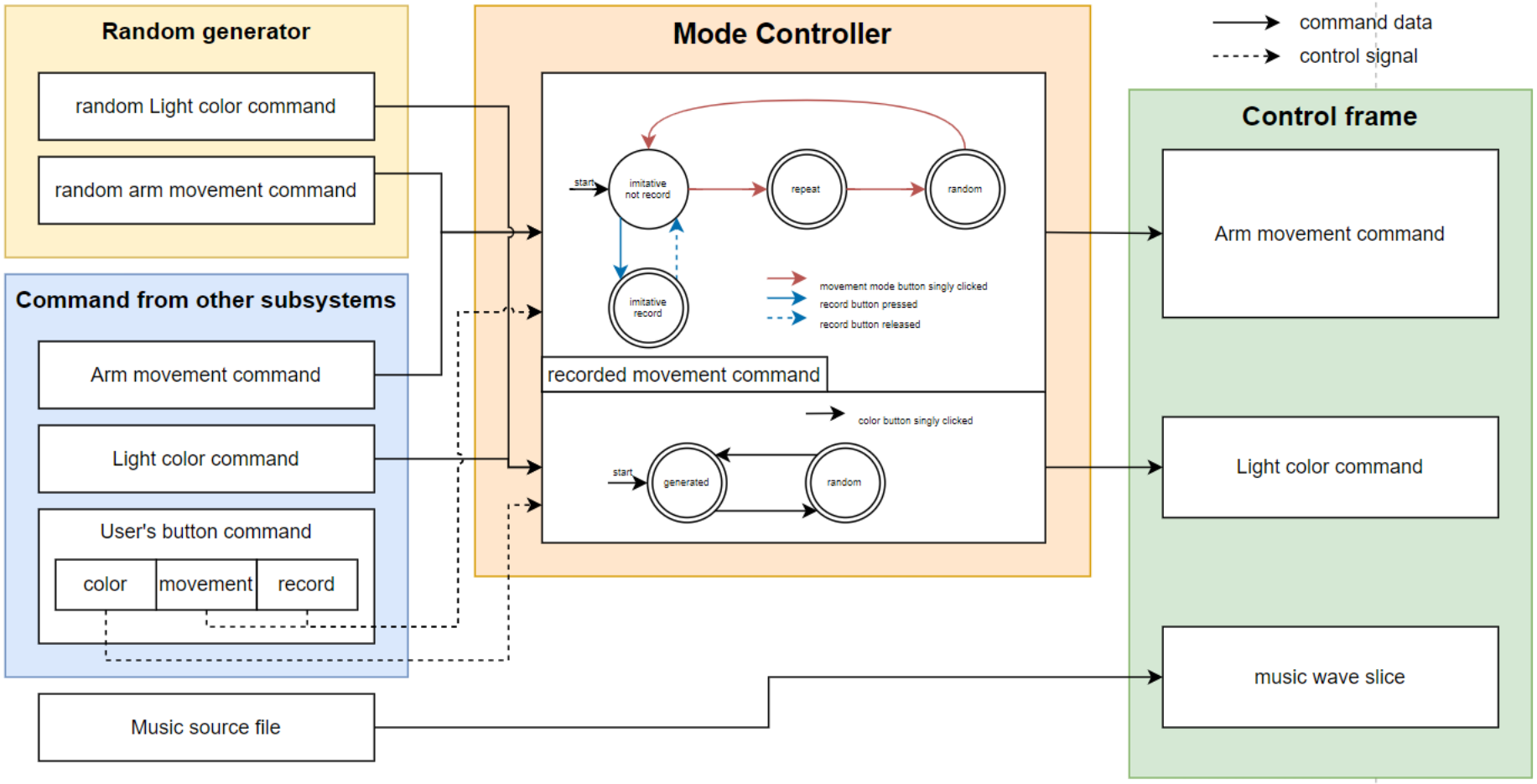


Control Subsystem – Design: FSM

- Finite State Machines (movement)
  - Movement states:
    - Mimic
    - Record
    - Repeat
    - Random
  - Light Color states:
    - Generated
    - Random



Control Subsystem – Design: Dataflow

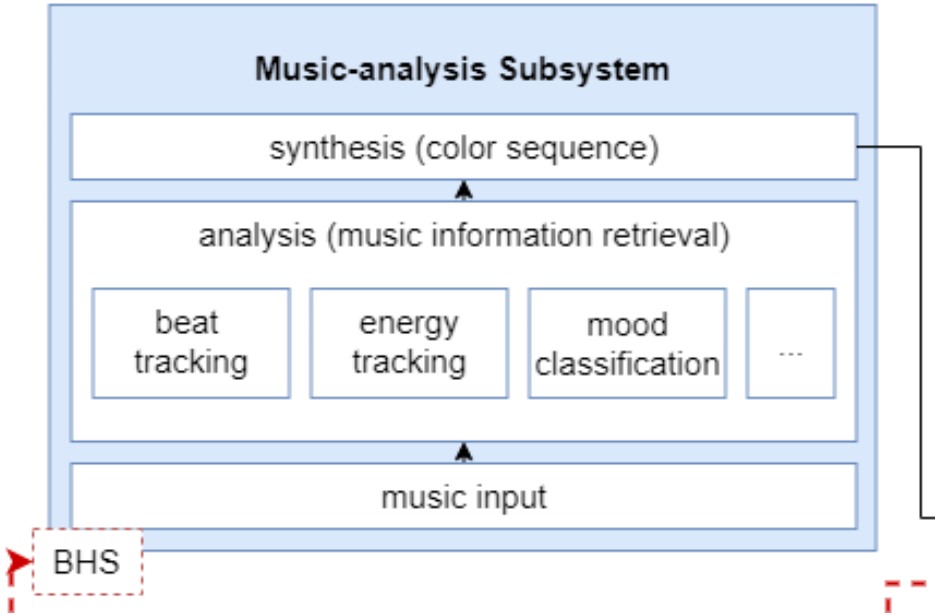


Music Analysis Subsystem

An Software Service that Generates Light Color Sequence

- Music Feature **Analysis**
  - Mel Spectrogram
  - Tempo, Beat Tracking
  - Energy Tracking
  - Emotion Recognition
- Color Sequence **Synthesis**
  - Heuristic Formula
  - HSV model
  - RGB model

Music → Color



Music Analysis – Analysis – R&V&Results

Requirements

- The change in color should be somehow align with the **latent feature of music.**

Verification & Results

- We extract features like: **tempo, beat sequence, energy and Mel spectrogram** to synthesize colors.
- The music features will be extracted beforehand. The requirements are met.

Modification

- Other than the basic music features, we add an advanced **emotion recognition** mechanism: a neural network trained from scratch. We achieved a accuracy of 70% on 4Q-emotion dataset.

Music Analysis – Synthesis – R&V&Results

Requirements

- The change in color should be somehow align with the **latent feature of music.**

Verification & Results

- We design a **heuristic synthesis formula** to generate color sequence.
- The music feature is mapped to color space using **HSV color model**.
- The time cost of overall computation is reduced to at most **2 seconds** (for normal pop music).

Modification

- The emotion recognized by the neural network (4 emotions) is involved. The Hue space in the HSV model is divided into 4 quadrants to express the emotion of the song.

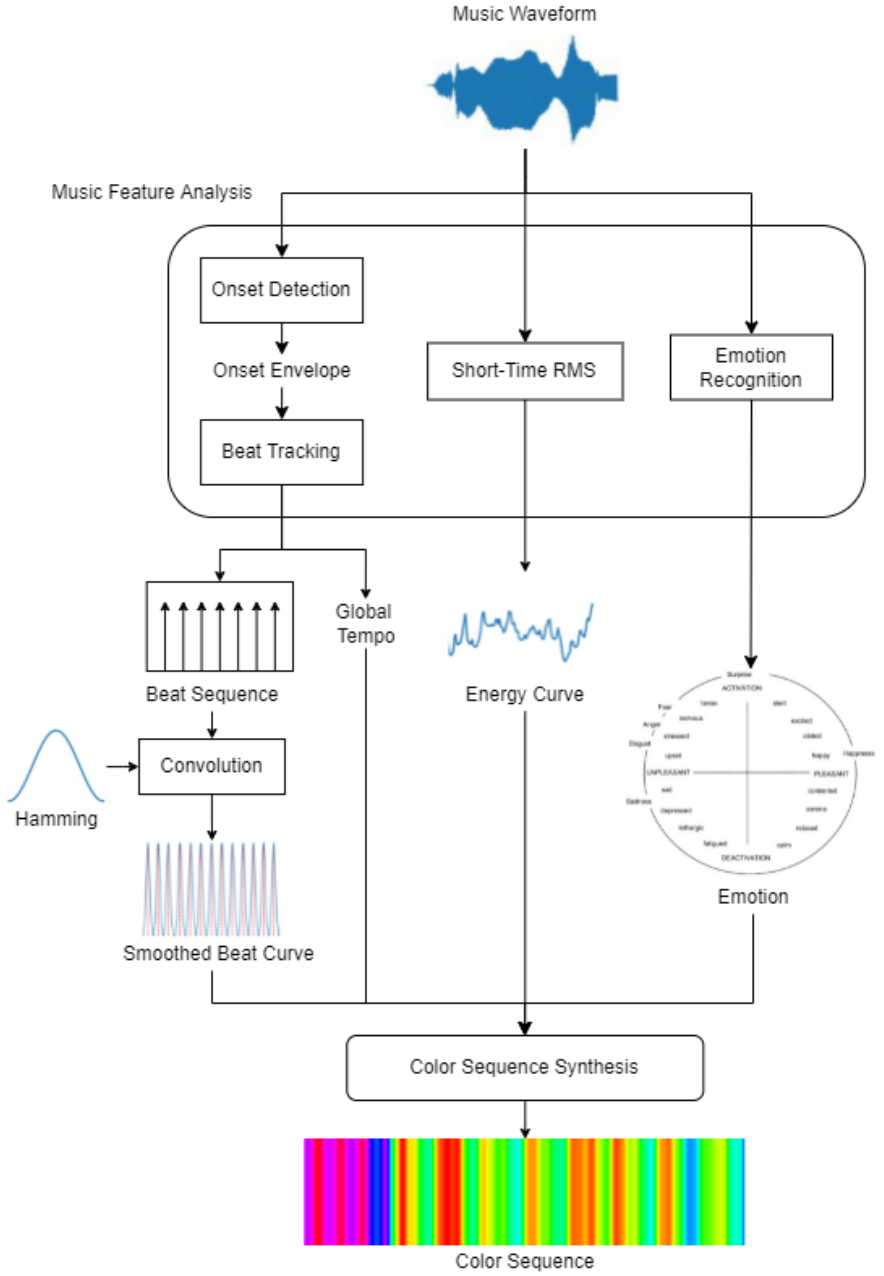
Music Analysis – Flow Chart

Heuristic Formula

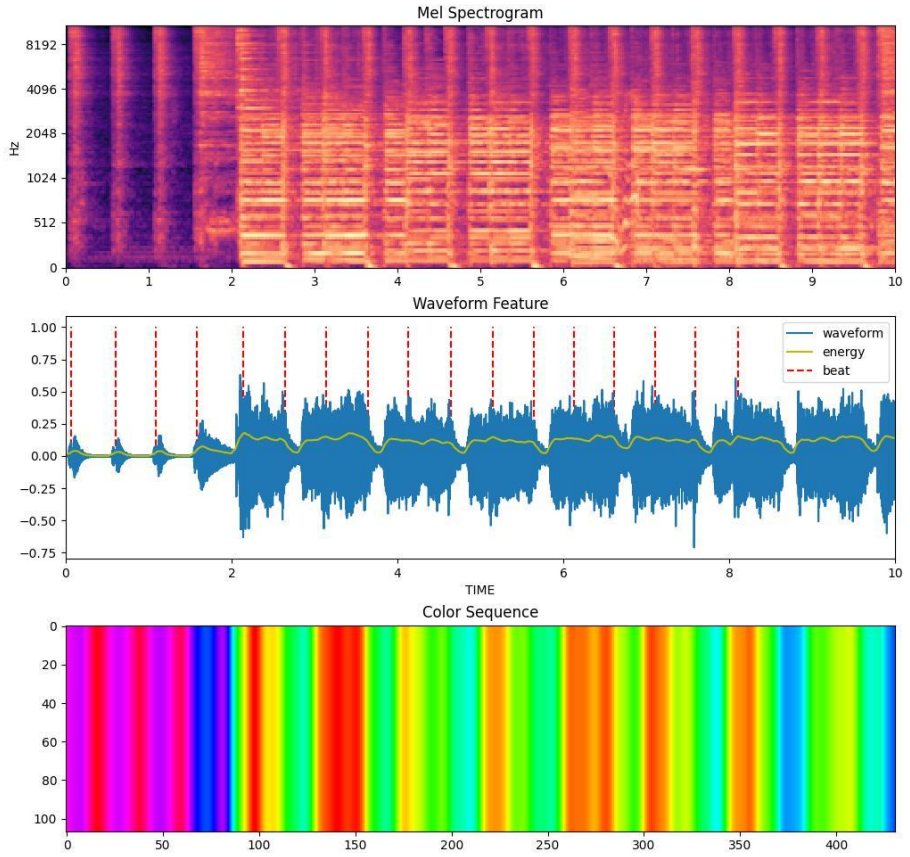
$$H[i] = E[i] + 0.6 \times \tilde{b}[i]$$

$$H[i] := \frac{H[i]}{\max(H[i])}$$

$$H[i] := (H[i] \times 240 + M \times 90 + 360) \mod 360$$



Music Analysis – Test





Music Analysis – Test

Time Cost Analysis

(Using timer library built in Python)

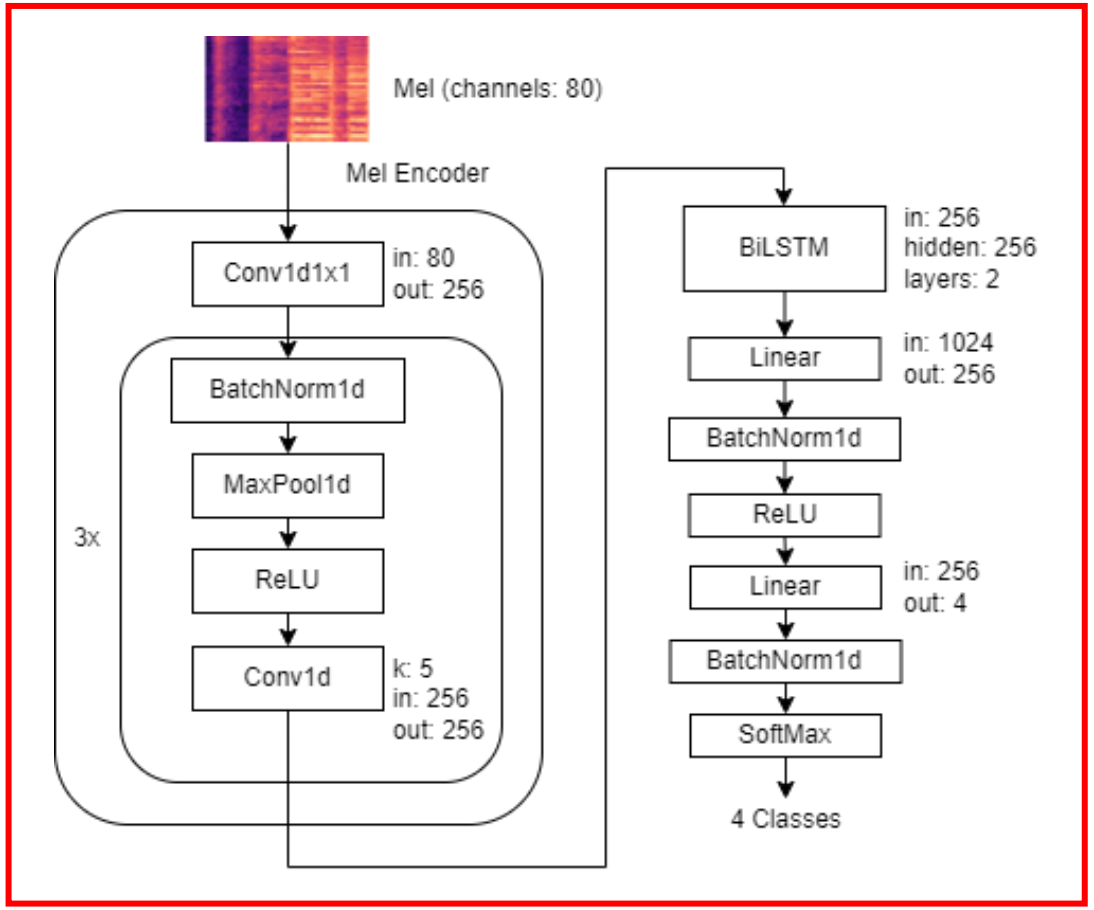
- loading: 7.12s
- beat tracking: 1.39s
- energy tracking: 0.16s
- emotion recognition: 0.40s
- color sequence mapping: 0.03s

The load time is the time of loading and decoding the MP3 file, which can be improved by switching to **streaming read**.

Music Analysis – Challenges

Emotion? Recognition?

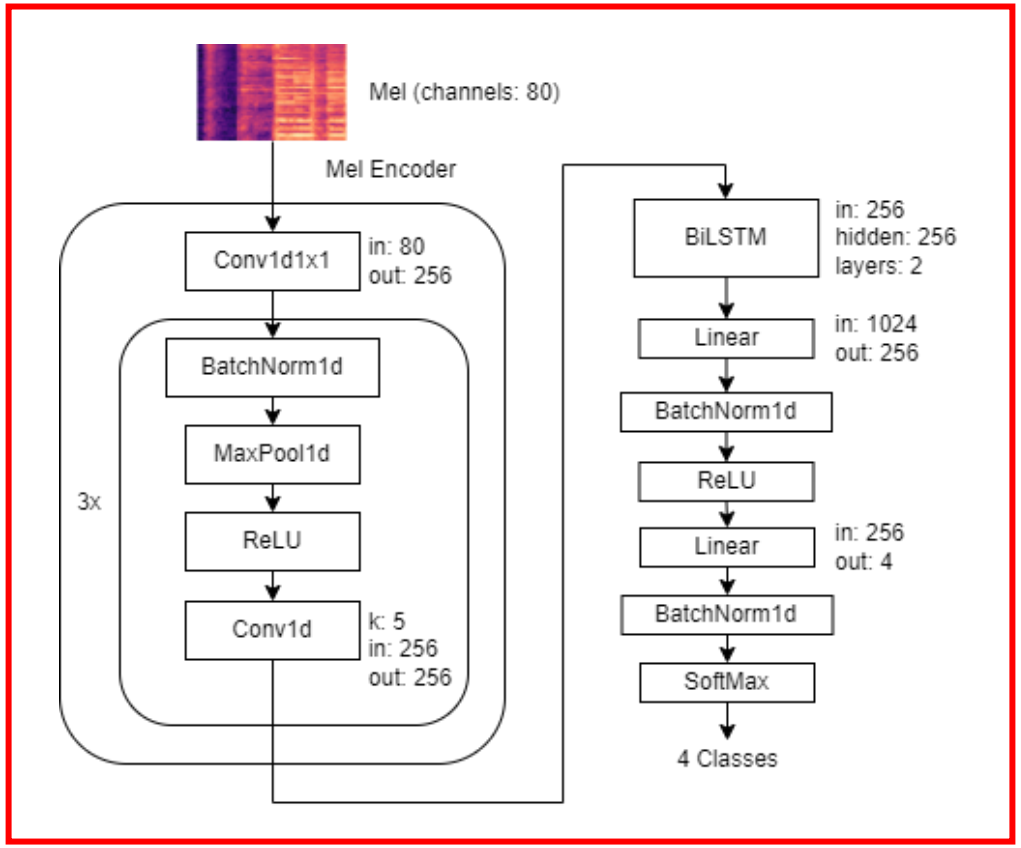
- We designed and trained this neural network on 4Q-emotion dataset (http://mir.dei.uc.pt/downloads.html)
- An accuracy of **70%** is achieved.
- The time cost of this neural network on a Intel gen 11 i5 CPU is below **0.5 seconds**, for a regular 4-min-long pop song.



Music Analysis – Challenges

The training of the neural network is **expensive** on my own PC (NVIDIA MX450). Initially, one epoch of training takes about **6 hours**.

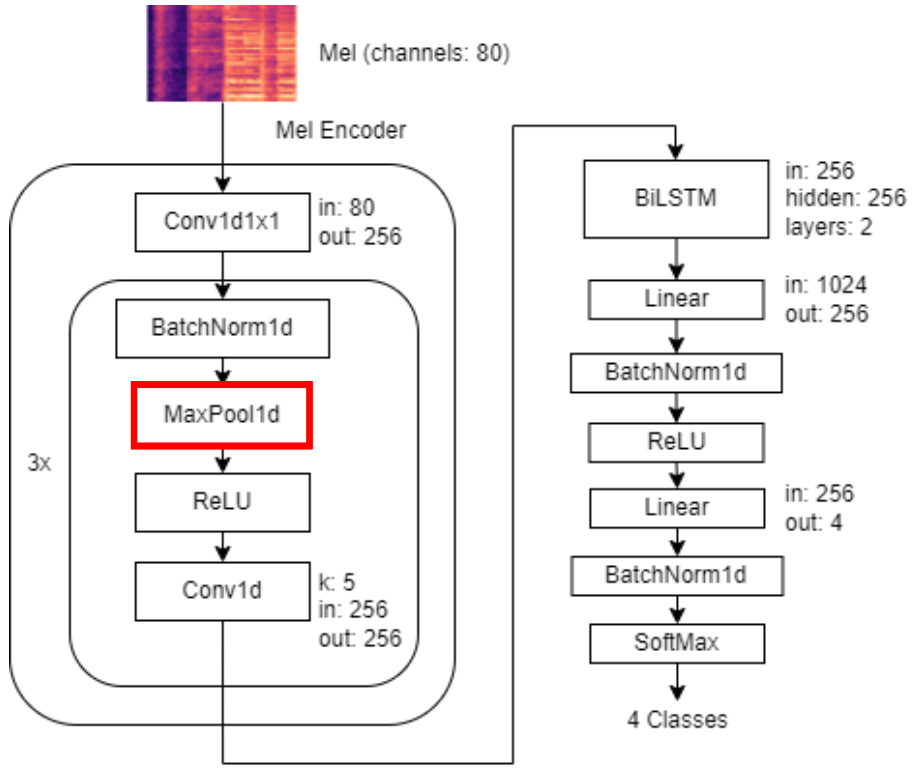
- We stacks multiple **MaxPool** layers to reduce the length of feature representations dramatically, so the time cost of RNN (LSTM) can be reduced.
- We add several **BatchNorm** layers to accelerate convergence procedure, so only a small number of training epochs is needed (about 20 epochs)



Music Analysis – Challenges

The training of the neural network is **expensive** on my own PC (NVIDIA MX450). Initially, one epoch of training takes about **6 hours**.

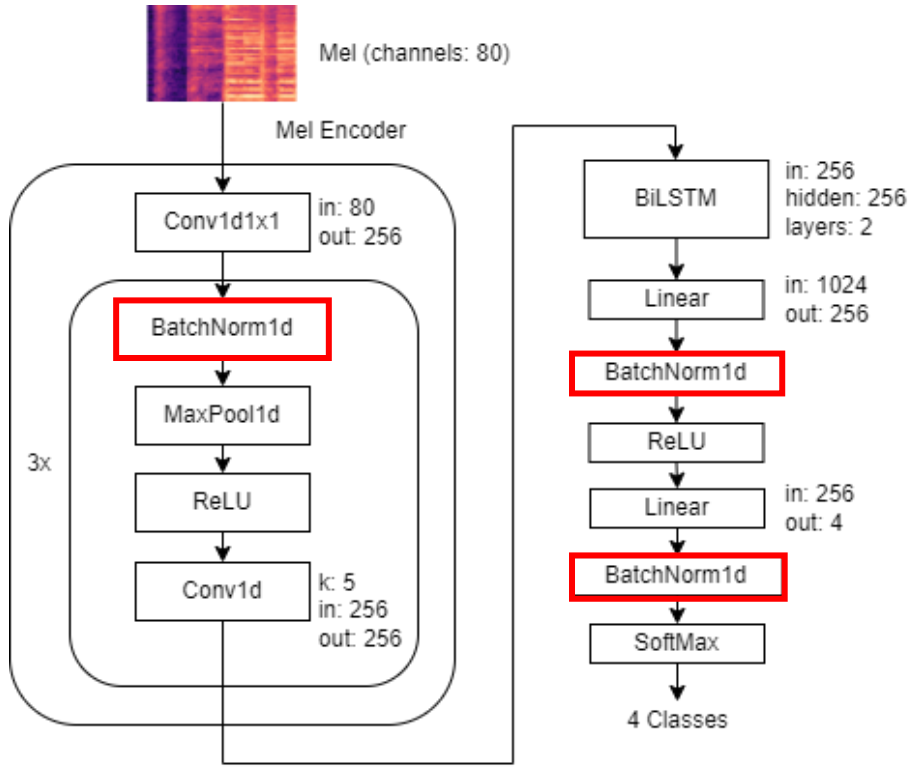
- We stacks multiple **MaxPool** layers to reduce the length of feature representations dramatically, so the time cost of RNN (LSTM) can be reduced.
- We add several **BatchNorm** layers to accelerate convergence procedure, so only a small number of training epochs is needed (about 20 epochs)



Music Analysis – Challenges

The training of the neural network is **expensive** on my own PC (NVIDIA MX450). Initially, one epoch of training takes about **6 hours**.

- We stacks multiple **MaxPool** layers to reduce the length of feature representations dramatically, so the time cost of RNN (LSTM) can be reduced.
- We add several **BatchNorm** layers to accelerate convergence procedure, so only a small number of training epochs is needed (about 20 epochs)



Lighting Array Subsystem

Control

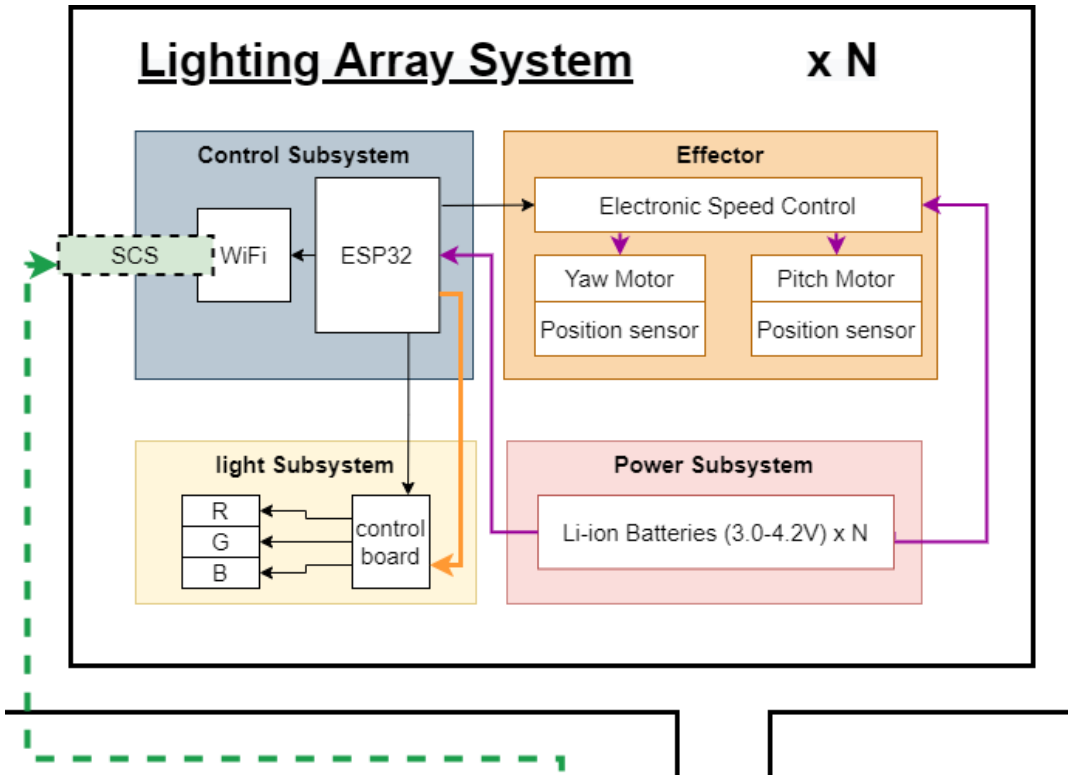
- ESP32
  - MCPWM generator - Servo
  - RMT peripheral - LED
  - WiFi socket - communication

Robot Arm / Effector / Manipulator

- Servo DS3218

Light Strip

- LED WS2812



Lighting Array – Robot Arm – R&V&Results

Requirements

- Given a PWM signal, the pitch servo output angle should be **locked**.
- The **delay** of the effectors should be minimized
- Use **PID** closed-loop control to command the RM6623 step motor for yaw axis.

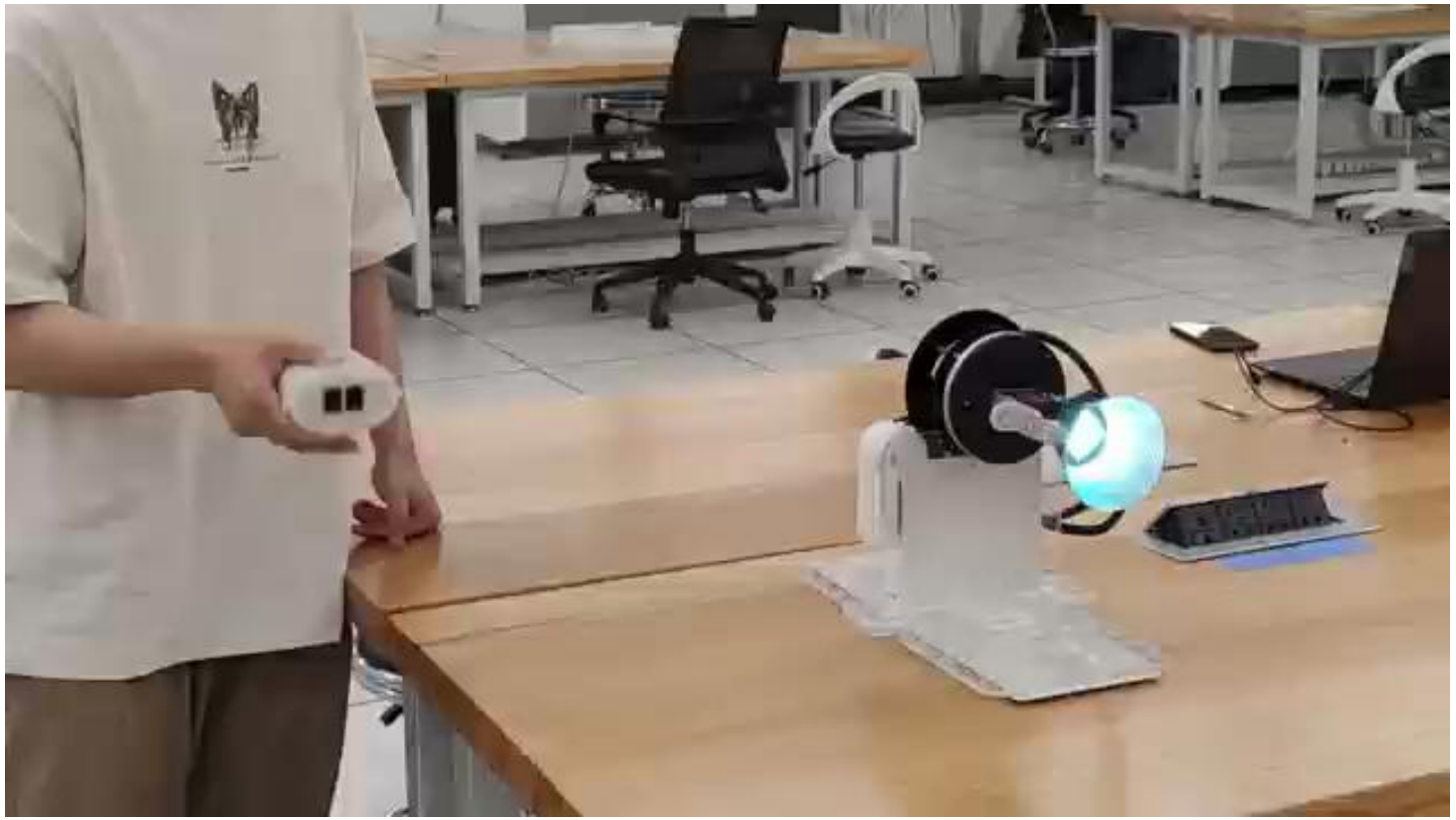
Verification & Results

- We use **20KG** servo to make sure the angle is locked stably.
- We use **MCPWM** peripheral built in ESP32 to generate stable PWM signal
- The **error** of control is reduced to 0.18° (tested by sending small angle commands)

Modification

- We build our robot arm along **pitch and roll** axis, instead of **pitch and yaw** axis, to avoid nonaxiality and instability.
- We use another **servo** for roll(yaw) axis instead of **RM6623 step motor**, because we cannot purchase CAN bus transceiver during pandemic.

Lighting Array – Robot Arm – Tests



Lighting Array – Light Strip – R&V&Results

Requirements

- Implement back connection **protection** to protect LED strip.
- Implement **carrier modulation** to generate signal waveform to generate linearly varying color sequence.
- It should be compatible with the **DMX512** lighting control protocol.

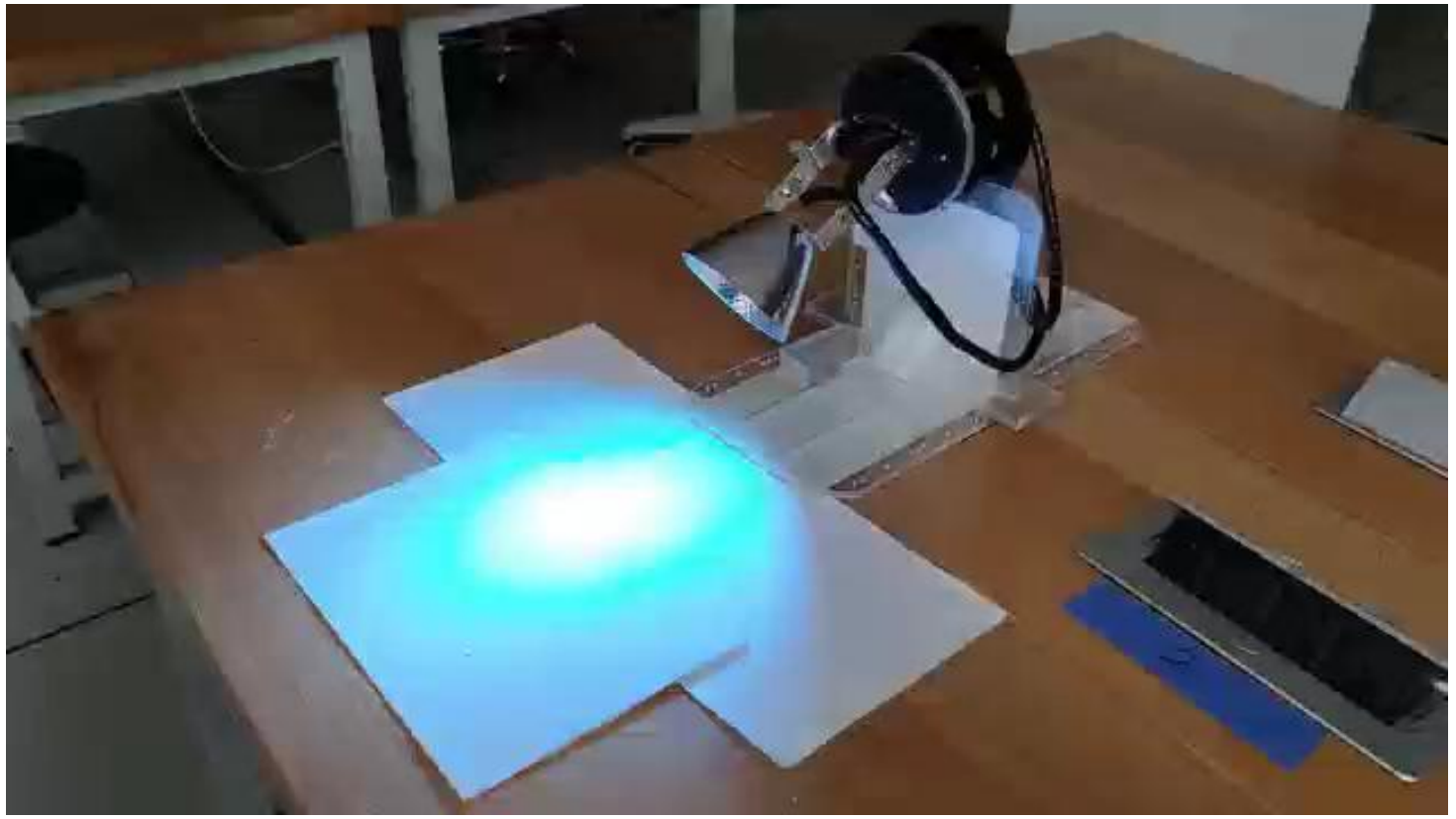
Verification & Results

- We use the Vout pin of ESP32 to power the LED strip, and the **power regulator** built in ESP32 protects the IC.
- We use **RMT (Remote Control)** peripheral built in ESP32 to implement the carrier modulation.
- The DMX512 protocol translator is implemented.

Modification

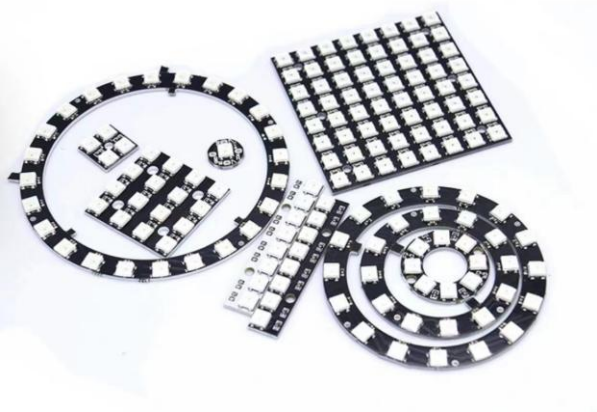
- We use the RMT peripheral to generate the waveform, instead of directly operating on the GPIO pin.
- “Array” means more than one lights, but because of the lacks of parts during pandemic, we only built one (more lights can be duplicated).

Lighting Array – Light Strip – Tests



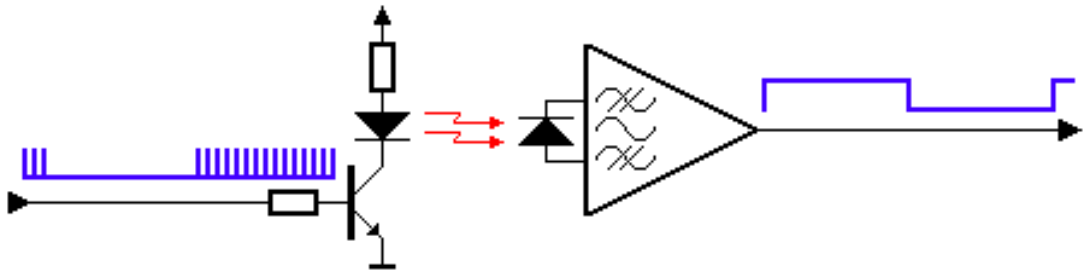
Lighting Array – Challenges

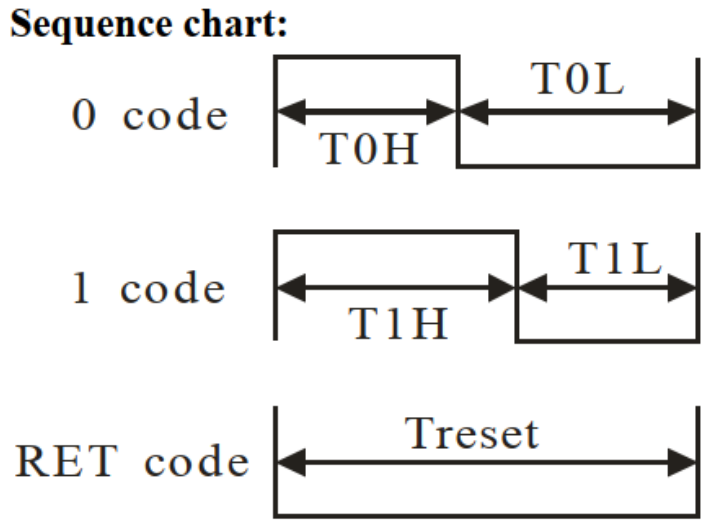
It's difficult to generate control signal waveforms through pure ESP32 programming.



Solution:

- Use the hardware peripheral built in RMT (**Remote Control**) module of ESP32 to implement the carrier modulation.





Mechanical Design

Two axis manipulator

- Position and orientation
  - Forward and inverse kinematics
- Moment of inertia and torque
  - Dynamics

Assembly

- Design for assembly
- Design for assembly of circuit and electronic element

High level requirement

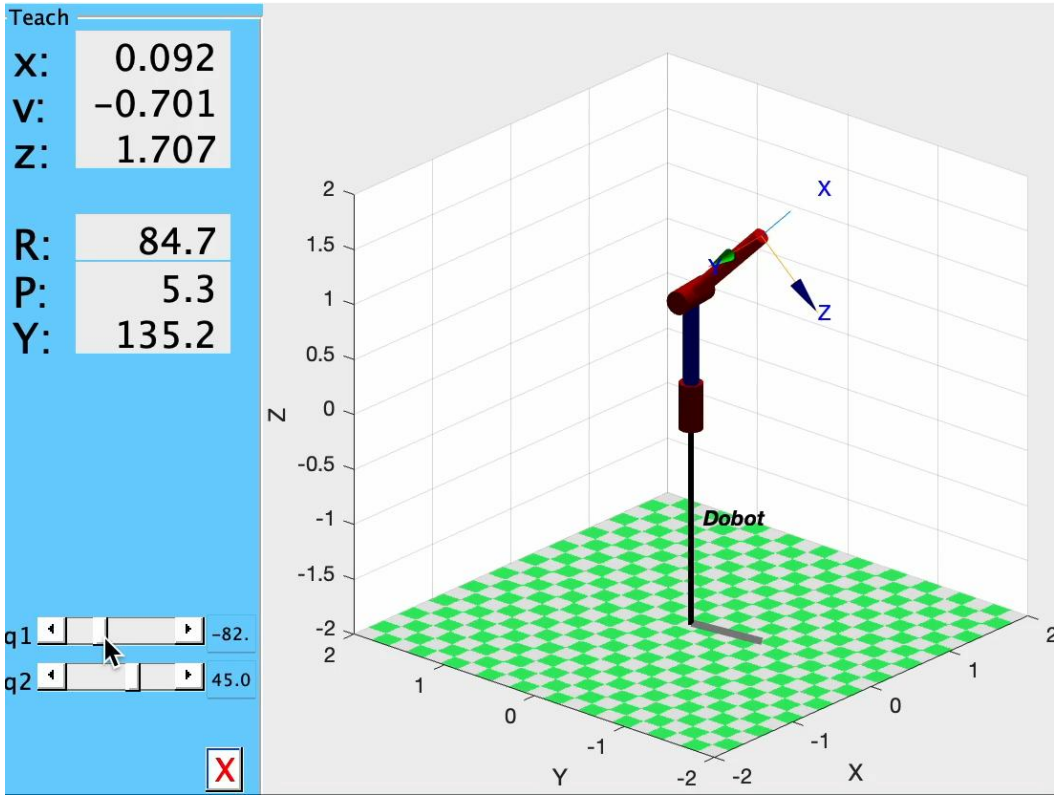
- Design for shape
- User friendly

Mechanical Design – Challenges - Position and orientation

The measured position and **orientation** do not correspond exactly to the servos.

Solution:

- Use Matlab to simulate the 2-axis manipulator
- The rotation angle of the servos to reach our desired **orientation** is obtained through inverse kinematics calculation



i	θ_i	d_i	a_{i-1}	α_{i-1}	rotation/translation
1	θ_1	212mm	0	$-\frac{\pi}{2}$	0
2	θ_2	0	90mm	0	0

Mechanical Design - Design for assembly

Assembly

- **Be standardized** and can be **easily assembled**

Disassembly

- Many **embedded structures** are used to ensure the detac hable mechanical structure

Material is limited

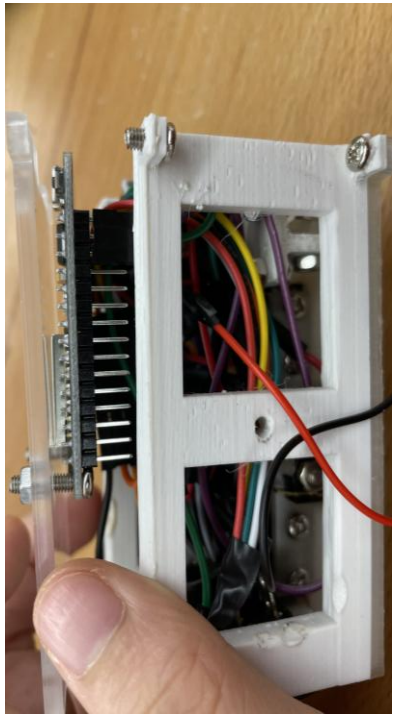
- Printed by 3D printing and laser cutting, plus standard parts



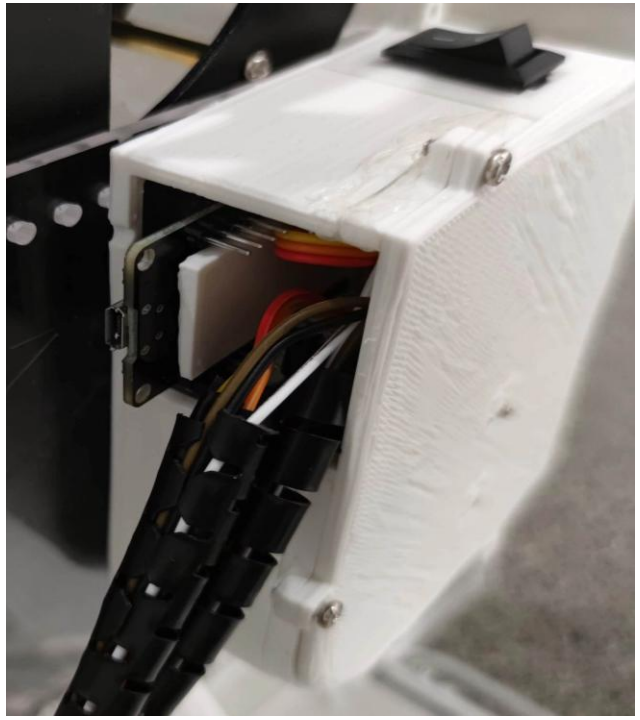
Mechanical Design - Design for assembly of circuit and electronic element

Wiring

- All wiring is packaged **inside the boxes**
- Each electronic component is fixed in a specific position
- with buttons and interfaces exposed, making it easy to input programs
- The outside of the wire is protected by an insulating layer







Mechanical Design - User friendly– high level requirement

Portable

- be able to be contained in a **40L bag and weights 9.3 kg**
- Has a single-hand pull structure.



Concentration of light

- Through the arc structure and special inner grain, the lighting intensity is enhanced and the light is concentrated



Appearance

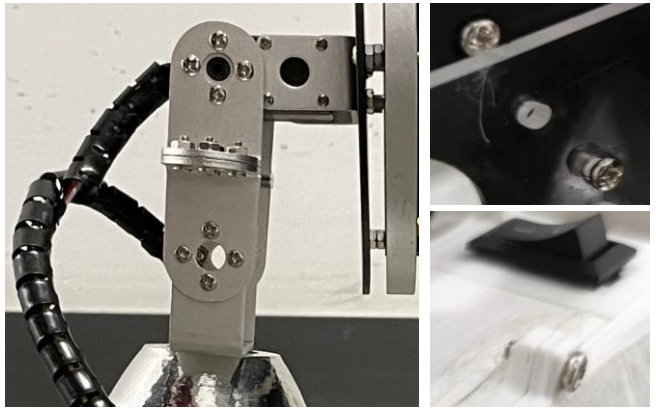
- All design use rounded corner and chamfer structure to ensure beautiful appearance and user safety



Mechanical Design - User friendly– high level requirement

Easy to maintain

- Use screws and standard parts for assembly
- After the failure of the user can be convenient for maintenance and view



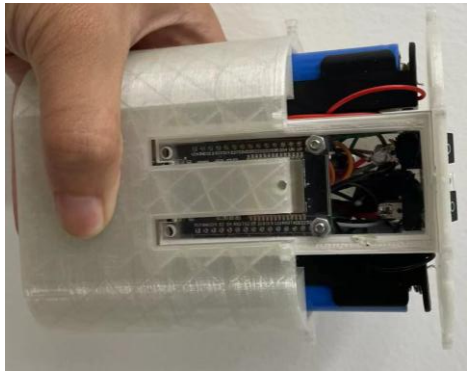
The switches distance

- The distance between Joystick switches is consistent with the user's **finger distance** to ensure user compatibility and comfort



Installation flexibility

- The battery are easily to be replaced without dismantling the entire unit
- The endurance of the robotics is **4.5 hours** under continuous use





3 Future Work

Future Work - Hardware

- The **workspace** of the robot arm may be limited. It's acceptable if the reachable angle meets the stage/room's coverage
- Reduce **response time** of servos
- Make the chassis rotate continuously 360 degrees and use **electric slip ring** to solve the winding problem
- Lighting with better lenses for more **focused lighting**
- Find the **lightest weight** that won't make the arm wobble -- making it easier to move
- Make arrays of lights that can be controlled with two hands
- ……

Future Work - Software

- The subsystems can be further **decomposed** to adapt requirements more flexibly
- The parallel program model of central server can be optimized to save computer resource
- The music analysis can be made in **real-time**, using streaming read
- A front-end App can be developed, supporting switching songs, adding new songs, creating song list
- ……



4 Ethics and Safety

Ethics

- Ethics of our product
- Ethics of research and development
  - No copy
  - No plagiarism
  - No existing solution
- Ethics sources
  - This project is bound and inspired by the Code of Ethics published by professional societies, such as IEEE and ACM.

Safety

- Production safety
- Electricity safety
- Mechanical structure safety
- Material processing safety

References

references

[1] J. Primrose, *Stage lighting system*, [Online], https://www.theatrecrafts.com/pages/home/topics/teaching/lighting/system/ Accessed March 23, 2022.

[2] USITT, *Dmx protocol basics*, [Online], http://www.dmx-512.com/dmx-protocol/dmx-protocol-basics/ Accessed March 23, 2022.

[3] Worldsemi, *Ws2812 datasheet*, [Online], https://d2j2m4p6r3pg95.cloudfront.net/module_files/led-cube/assets/datasheets/WS2812B.pdf Accessed March 27, 2022.

[4] D. P. Ellis, "Beat tracking by dynamic programming," *Journal of New Music Research*, vol. 36, no. 1, pp. 51–60, 2007.

[5] Wikipedia, *Mel_scale*, [Online], https://en.wikipedia.org/wiki/Mel_scale Accessed March 23, 2022.

[6] LDT, *Librosa.beat.beat$_{t}rack$*, [Online], https://librosa.org/doc/latest/generated/librosa.beat.beat_track.html#librosa.beat.beat_track Accessed March 23, 2022.

[7] A. R. Smith, "Color gamut transform pairs," in *SIGGRAPH '78*, 1978, pp. 12–19.

[8] Wiki, *Coordinate system*, [Online], https://en.wikipedia.org/wiki/Coordinate_system Accessed March 23, 2022.

[9] espressif, *Esp32-wroom-32 datasheet*, [Online], https://www.espressif.com/zh-hans/support/documents/technical-documents Accessed March 23, 2022.

[10] Intel-Corporation, *Intel® wi-fi 6 ax200 cyclone peak 2 product brief*, [Online], https://www.intel.com/content/dam/www/public/us/en/documents/product-briefs/wi-fi-6-ax200-module-brief.pdf Accessed March 24, 2022.

[11] MicroPython, *Micropython esp32 firmware*, [Online], https://micropython.org/download/esp32/.

[12] P. S. Damien P. George and contributors, *Quick reference for the esp32*, [Online], http://docs.micropython.org/en/latest/esp32/quickref.html Accessed March 23, 2022.

[13] espressif, *Esp32 datasheet*, [Online], https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf Accessed March 23, 2022.

[14] G. Welch, G. Bishop, *et al.*, "An introduction to the kalman filter," 1995.

[15] IEEE. ""IEEE Code of Ethics"." (2016), [Online]. Available: https://www.ieee.org/about/corporate/governance/p7-8.html (visited on 02/08/2020).

[16] OSHA. ""Occupational Safety and Health Administration"." (2016), [Online]. Available: https://www.osha.gov/ (visited on 02/08/2020).

[17] FCC. ""Federal Communications Commissions"." (2016), [Online]. Available: https://www.fcc.gov/ (visited on 03/23/2022).


Thanks for your listening